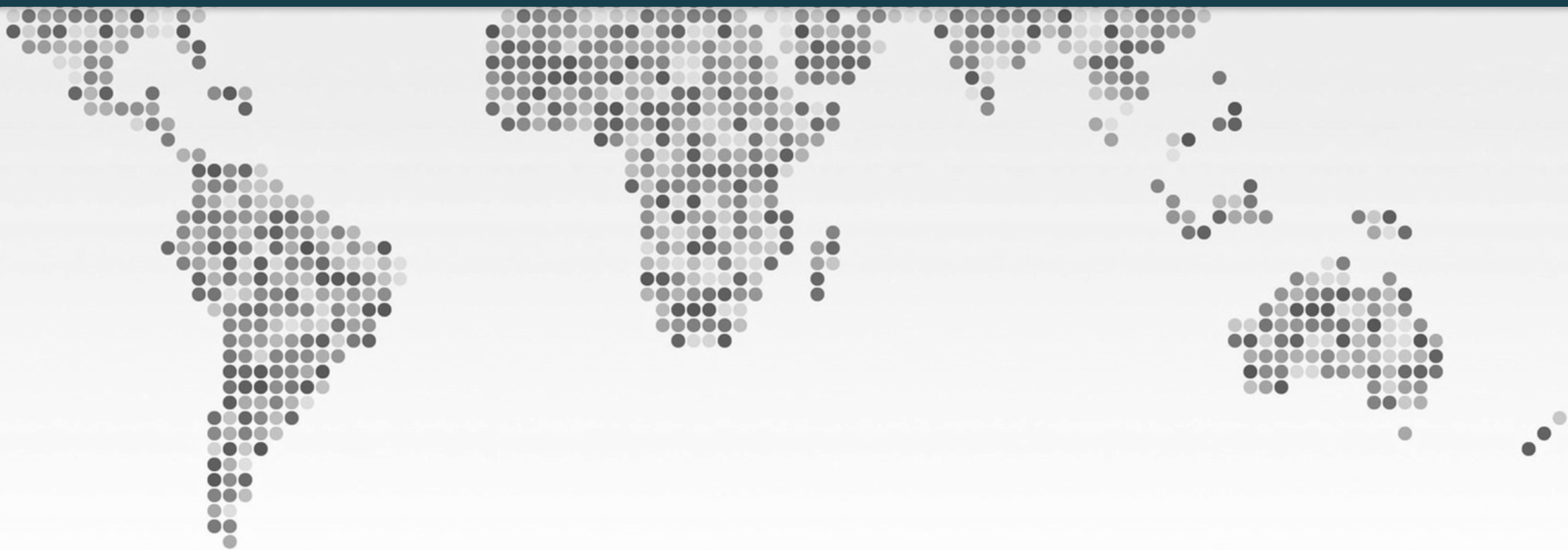


Software Engineering for Artificial Intelligence

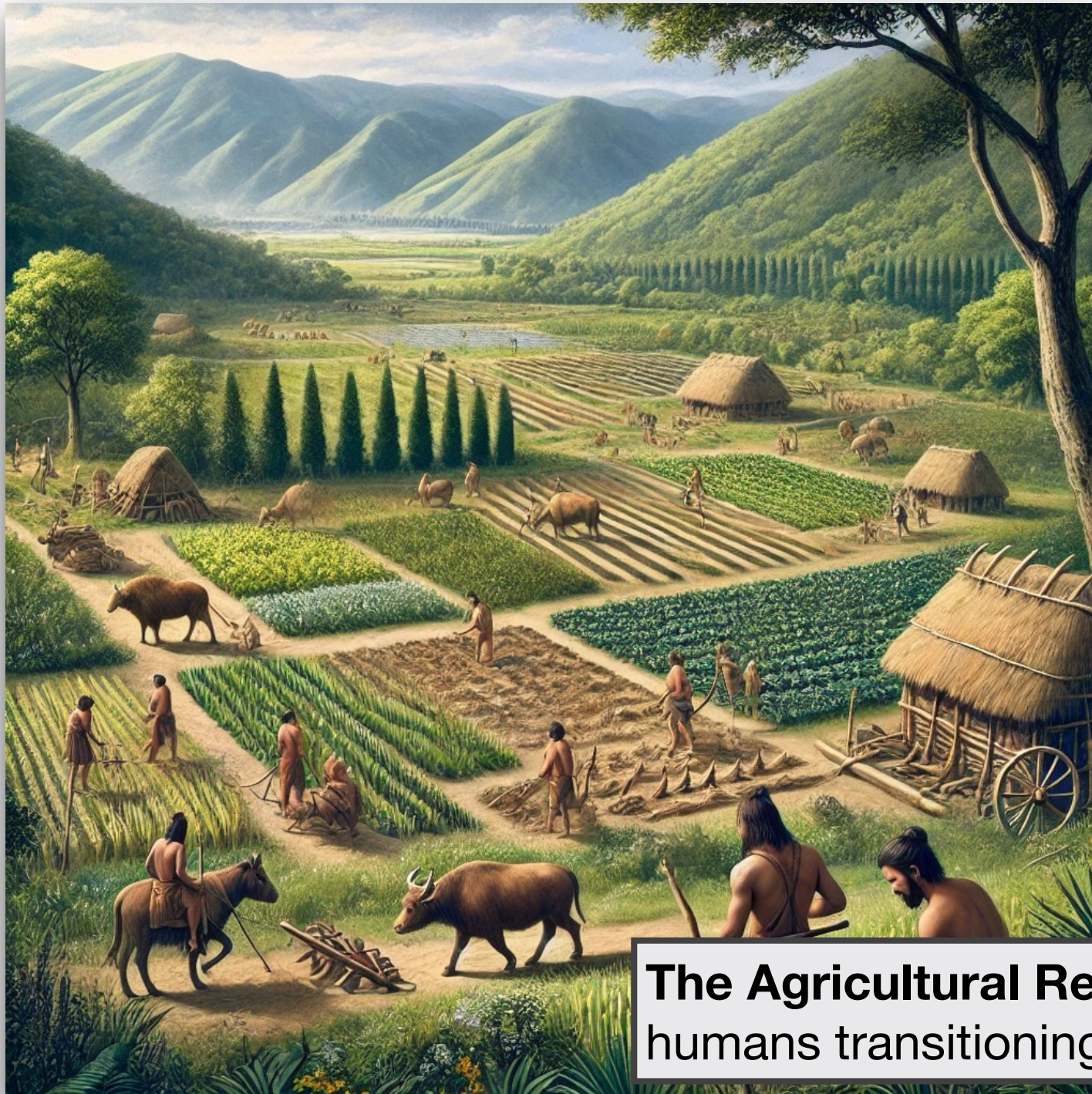
Configuring Artificial Intelligence Solutions III



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Whether you are passionate about history or not, there are moments that have forever shaped society, radically changing the way we think, act, and approach life.



The Agricultural Revolution (~10,000 BCE): Early humans transitioning from hunting to farming.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Whether you are passionate about history or not, there are moments that have forever shaped society, radically changing the way we think, act, and approach life.



World War II (1939-1945 / 20th Century): Wartime innovation leading to early computing.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Whether you are passionate about history or not, there are moments that have forever shaped society, radically changing the way we think, act, and approach life.



9/11 (9/11/2001): 9/11 reshaped global politics, security policies, and societal perceptions of security.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

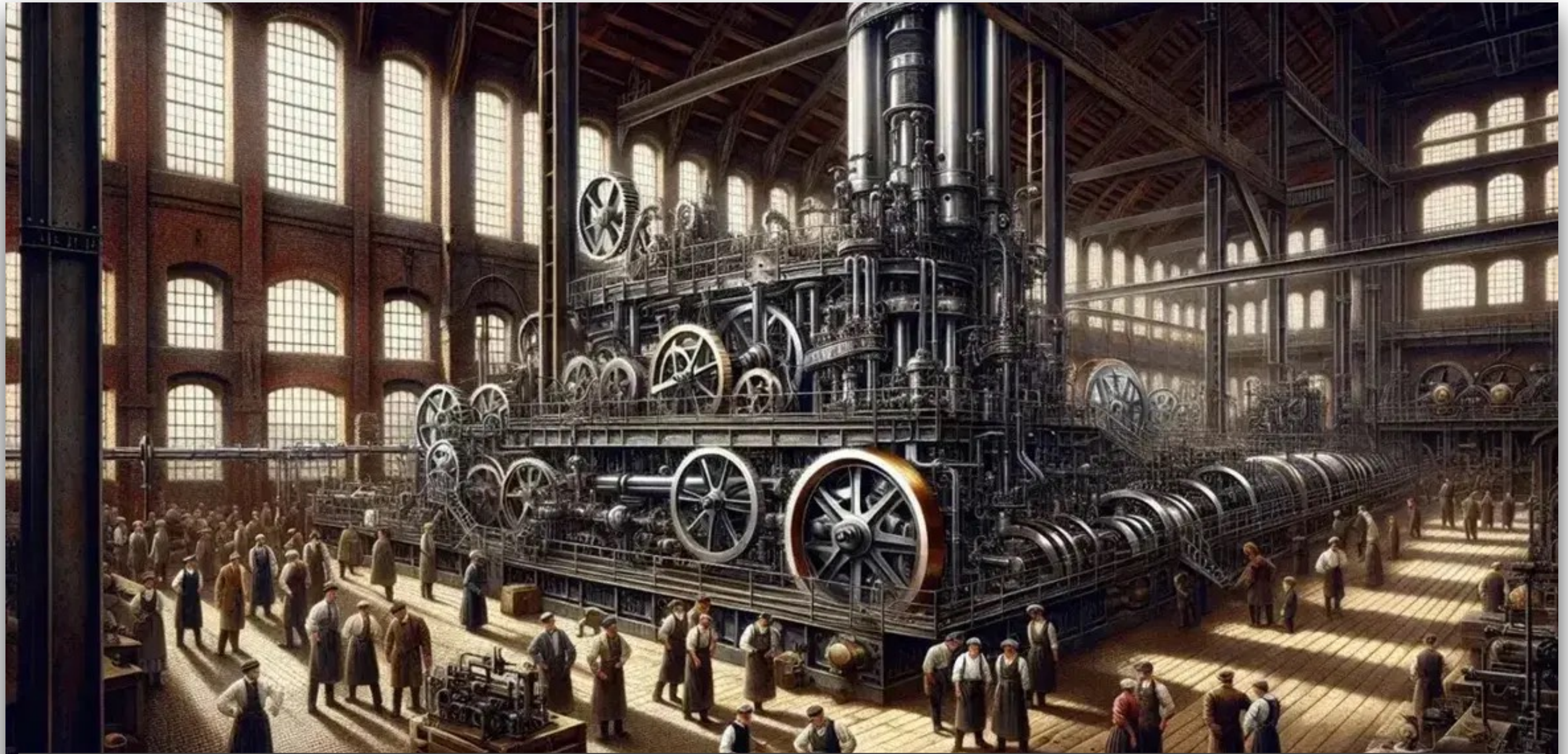
Whether you are passionate about history or not, there are moments that have forever shaped society, radically changing the way we think, act, and approach life.



The AI & Digitalization Era (Now): A futuristic world driven by artificial intelligence.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

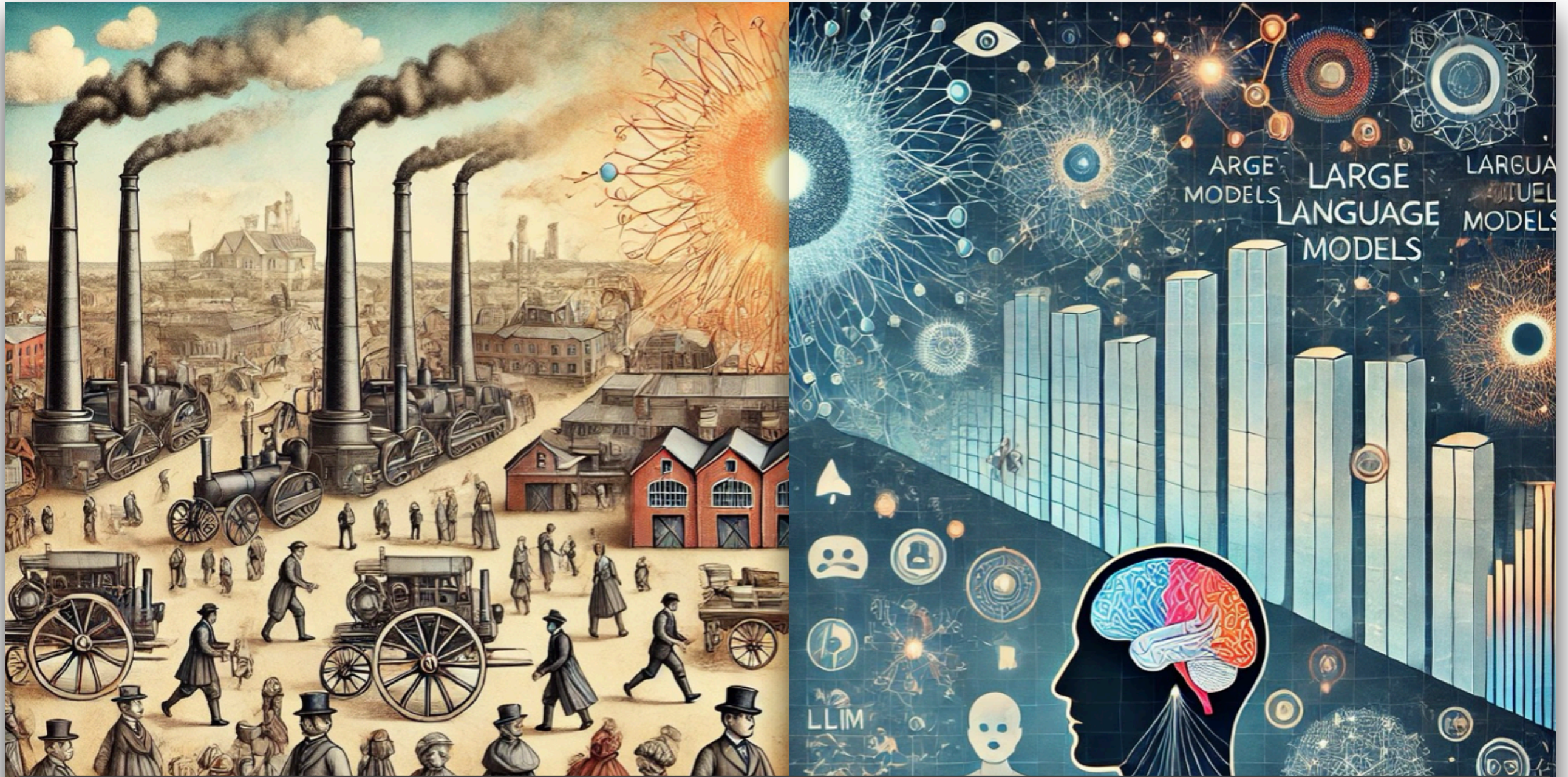
Configuring Artificial Intelligence Solutions: Large Language Models



The Industrial Revolution was a period of economic, technological, and social transformations that began in the 18th century. It was characterized by the introduction of machines, large-scale production, and the shift from an agricultural economy to an industrial one, radically changing society and human labor.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models



Large Language Models (LLMs) represent the Industrial Revolution of computing: whether we like it or not, there is a clear **before** and **after** their introduction. Nothing is the same as before. How do you see yourself in 10 years?

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Parallels Between LLM and the Industrial Revolution

Transformation of Production.

- Industrial Revolution: Introduced large-scale production through mechanization, transforming the artisanal economy into an industrial one. Machines replaced or amplified manual labor, drastically increasing efficiency.
- LLM Revolution: LLMs, such as ChatGPT, are revolutionizing content production (textual, creative, technical), automating complex tasks like writing, programming, and decision support.

Configuring Artificial Intelligence Solutions: Large Language Models

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Parallels Between LLM and the Industrial Revolution

Impact on the Workforce.

- Industrial Revolution: Created new jobs in factories but also reduced the demand for *traditional* manual labor, leading to a significant shift in terms of skills required to perform a job.
- LLM Revolution: LLMs are making some professions less central (e.g., repetitive tasks or basic analysis), pushing workers to develop more creative skills or focus on managing and overseeing technology.

Configuring Artificial Intelligence Solutions: Large Language Models

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Parallels Between LLM and the Industrial Revolution

Democratization of Resources.

- Industrial Revolution: Mass production made previously expensive or exclusive goods more accessible to a wider audience.
- LLM Revolution: LLMs are democratizing access to advanced artificial intelligence tools, making knowledge, translations, technical assistance, and creative support available to a broad audience.

Configuring Artificial Intelligence Solutions: Large Language Models

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Parallels Between LLM and the Industrial Revolution

Change in Infrastructure.

- Industrial Revolution: Required the development of new infrastructures, such as factories, railways, and transportation systems.
- LLM Revolution: Requires massive digital infrastructures, including data centers, cloud computing, and high-performance processing systems.

Configuring Artificial Intelligence Solutions: Large Language Models

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Parallels Between LLM and the Industrial Revolution

Driving Innovations.

- Industrial Revolution: Innovations such as the steam engine, mechanical looms, and advancements in metallurgy drove the transformation.
- LLM Revolution: Neural model architectures (e.g., Transformers), large-scale data availability, and improvements in computational hardware (GPUs, TPUs) are the key innovations.

Configuring Artificial Intelligence Solutions: Large Language Models

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Parallels Between LLM and the Industrial Revolution

New Challenges.

- Industrial Revolution: Led to issues such as labor exploitation, environmental pollution, and economic inequalities.
- LLM Revolution: Is raising ethical and social concerns, including algorithmic biases, misinformation, copyright issues, and privacy implications.

Configuring Artificial Intelligence Solutions: Large Language Models

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Parallels Between LLM and the Industrial Revolution

Cultural Change.

- Industrial Revolution: Radically transformed society by introducing new urban lifestyles and accelerating scientific progress.
- LLM Revolution: Is changing how people learn, communicate, and create, influencing education, entertainment, and work.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Where it all began...

As we have seen throughout this course, machine learning has made significant progress in various areas, including classification, regression, and clustering.

However, many of these techniques perform well under a specific assumption: the **training** and **test data** come from the **same feature space** and follow the **same distribution**.

When the distribution changes, these statistical models need to be rebuilt from scratch using newly collected training data. This process can be **costly** or, in some cases, **even impossible** to carry out.

For example, consider a model that predicts umbrella sales based on rainfall data in a city. If the model was trained on data collected in a city with a rainy climate but is used in a city with a dry climate, the data distribution changes completely!

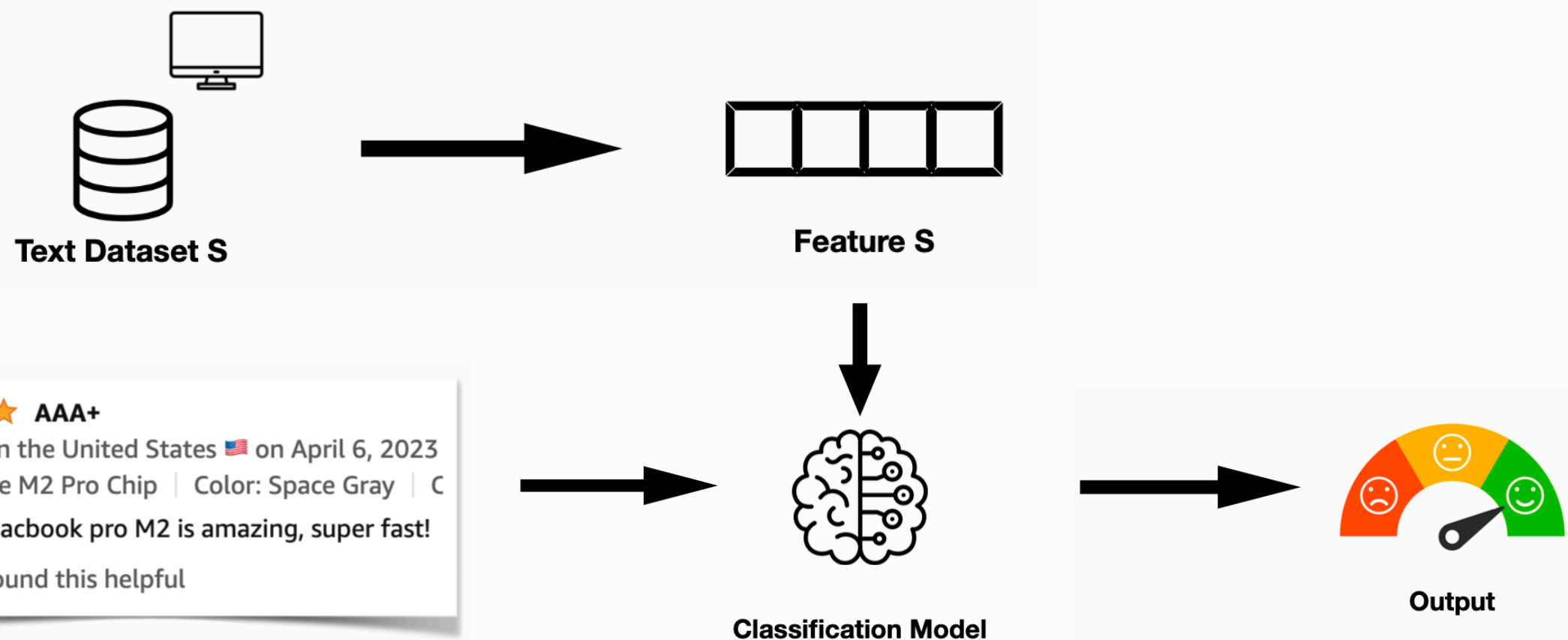
Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Where it all began...

Let's consider the example of sentiment analysis for product reviews.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Where it all began...

Let's consider the example of sentiment analysis for product reviews.

What happens if, instead of a review of a computer, the model receives a review of a smartphone? Will the performance remain the same?

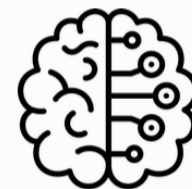
★★★★★ AAA+

Reviewed in the United States 🇺🇸 on April 6, 2023

Style: Apple M2 Pro Chip | Color: Space Gray | C

The new macbook pro M2 is amazing, super fast!

6 people found this helpful



Classification Model



Output

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Where it all began...

The model was trained on data related to computer reviews, not smartphone reviews. Therefore, there could be a decrease in performance if the received data differs significantly from the training set data.

To maintain good performance, we should design our solution in such a way that there is a sufficient amount of training data for each product we expect to encounter in the operational environment.

Or...

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Where it all began...

From Software Engineering, we can borrow some terms... for example, the concept of **reuse**.

Reuse: The practice of using a model already trained on a specific problem or domain as a foundation for solving another problem, reducing the need to build a model from scratch. This technique allows leveraging pre-existing knowledge learned by the model, saving time, computational resources, and data.

In a world driven by data, this seems to be an interesting solution... especially considering that many of the problems we aim to solve have already been tackled in the past, or at least addressed, using models and approaches that can be adapted or reused for similar contexts.

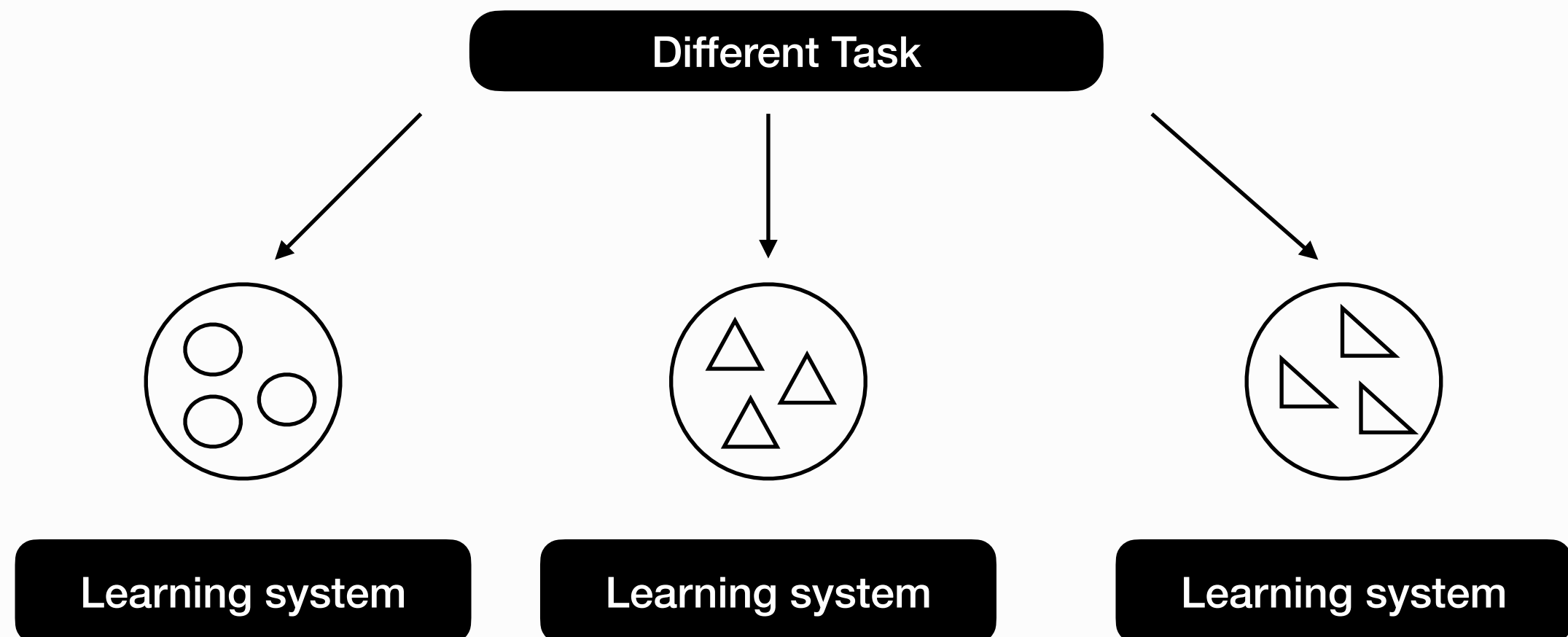
Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Where it all began...

This means transitioning from a situation like this...



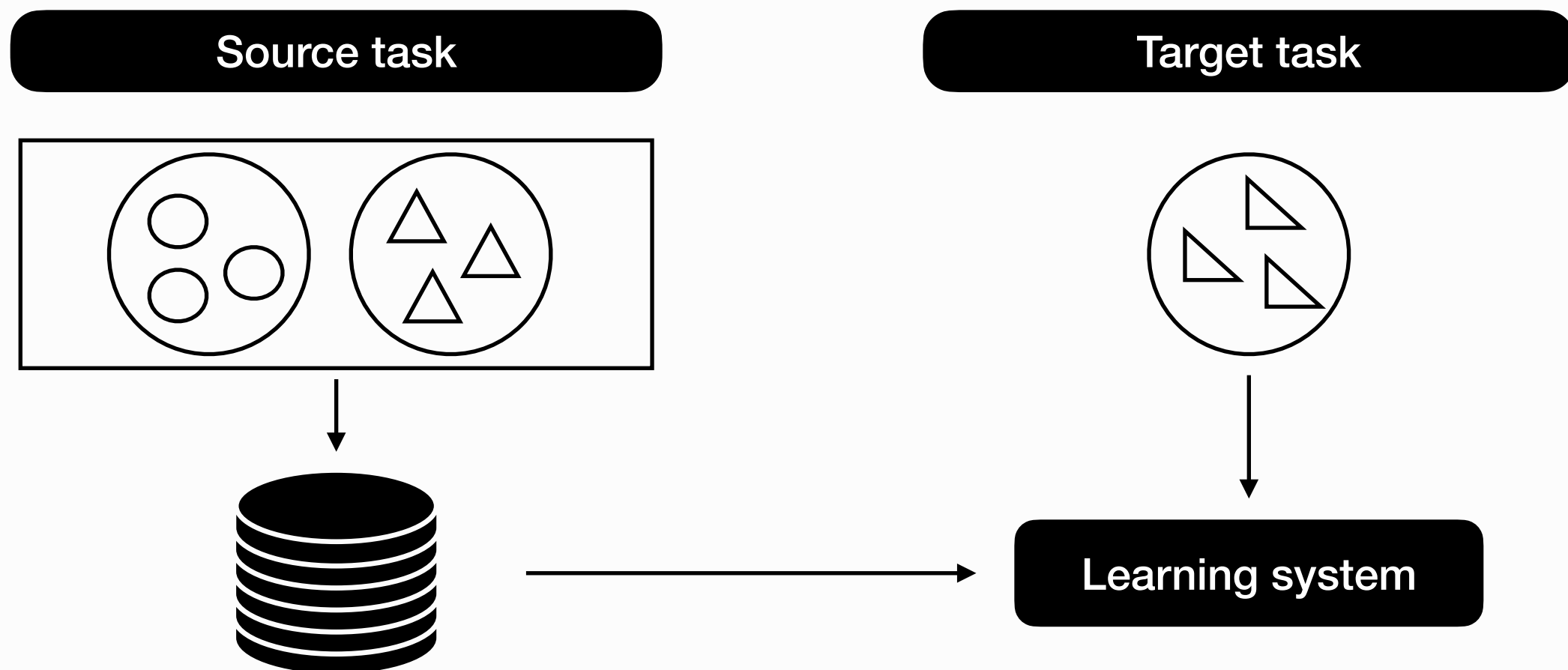
Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Where it all began...

To something like...

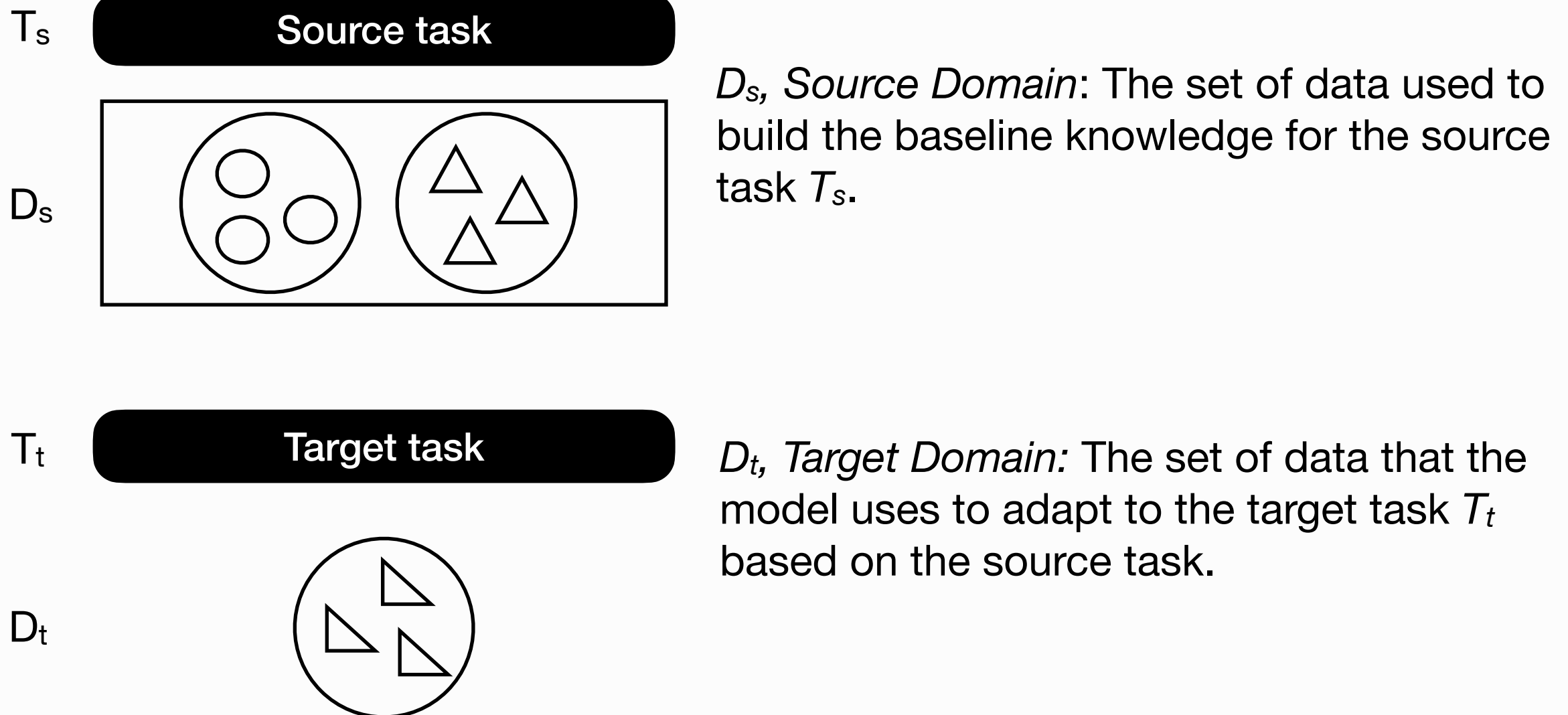


Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Where it all began...



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Where it all began...

Given D_s , D_t , T_s e T_t , we can define *transfer learning*.

Transfer Learning: A machine learning technique in which a pre-trained model on domain D_s for task T_s is reused, adapted, or fine-tuned to solve a new task T_t in domain D_t .

It is important to note that in transfer learning, the following property holds:
 $D_s \neq D_t$ OR $T_s \neq T_t$. It may seem trivial, but be careful: there's an **OR**, not an **AND**!

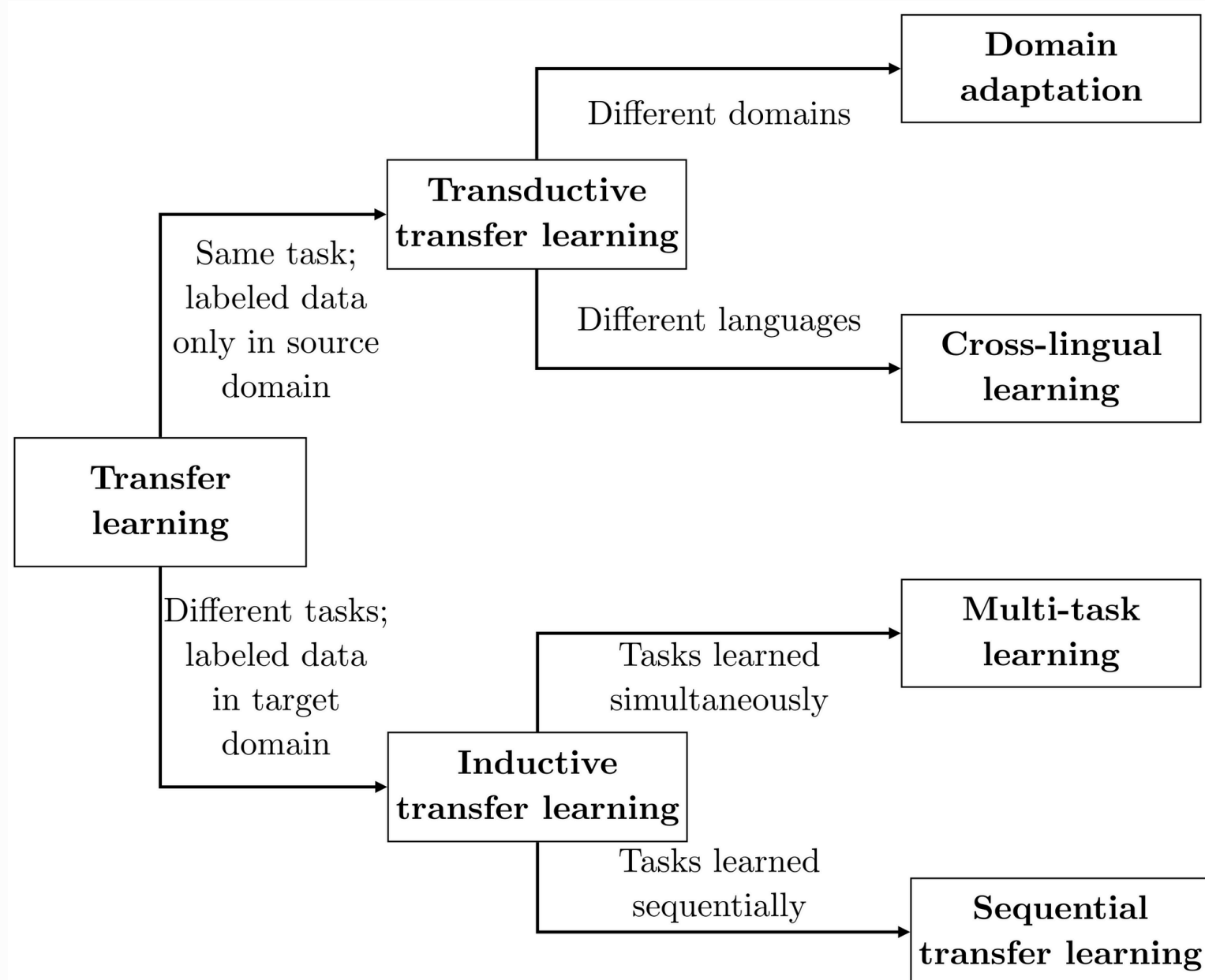
This means that we can reuse a model trained in the same domain by changing the required task or vary the domain to train a model for the same task.

In other terms, this property allows us to either apply a pre-trained model to a new task within the same domain or adapt it to a new domain while solving the same task, enabling more efficient learning and reducing the need for labeled data.

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Configuring Artificial Intelligence Solutions: Large Language Models

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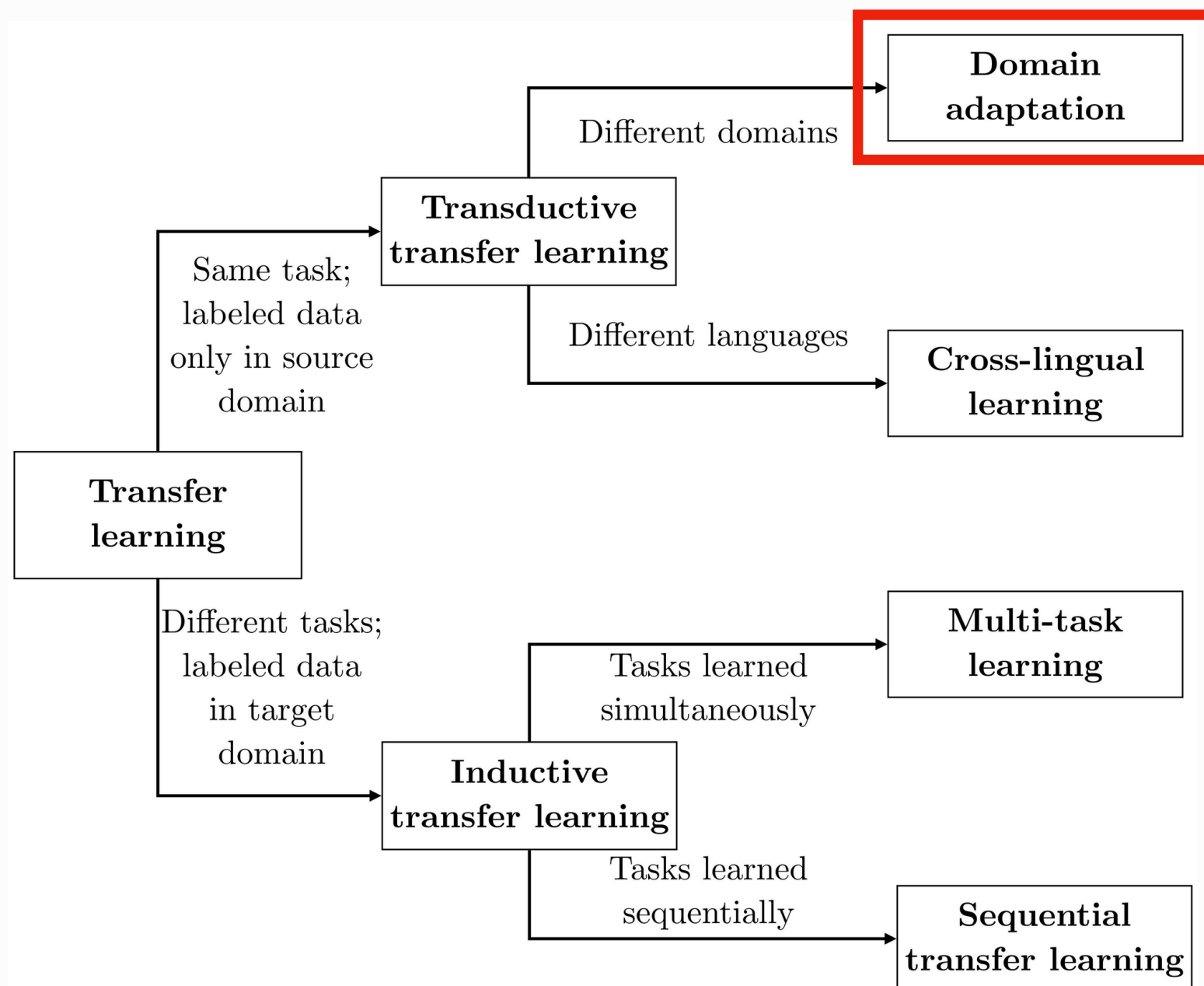


Because of the property, we may end up with various types of transfer learning types.

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Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.



Example

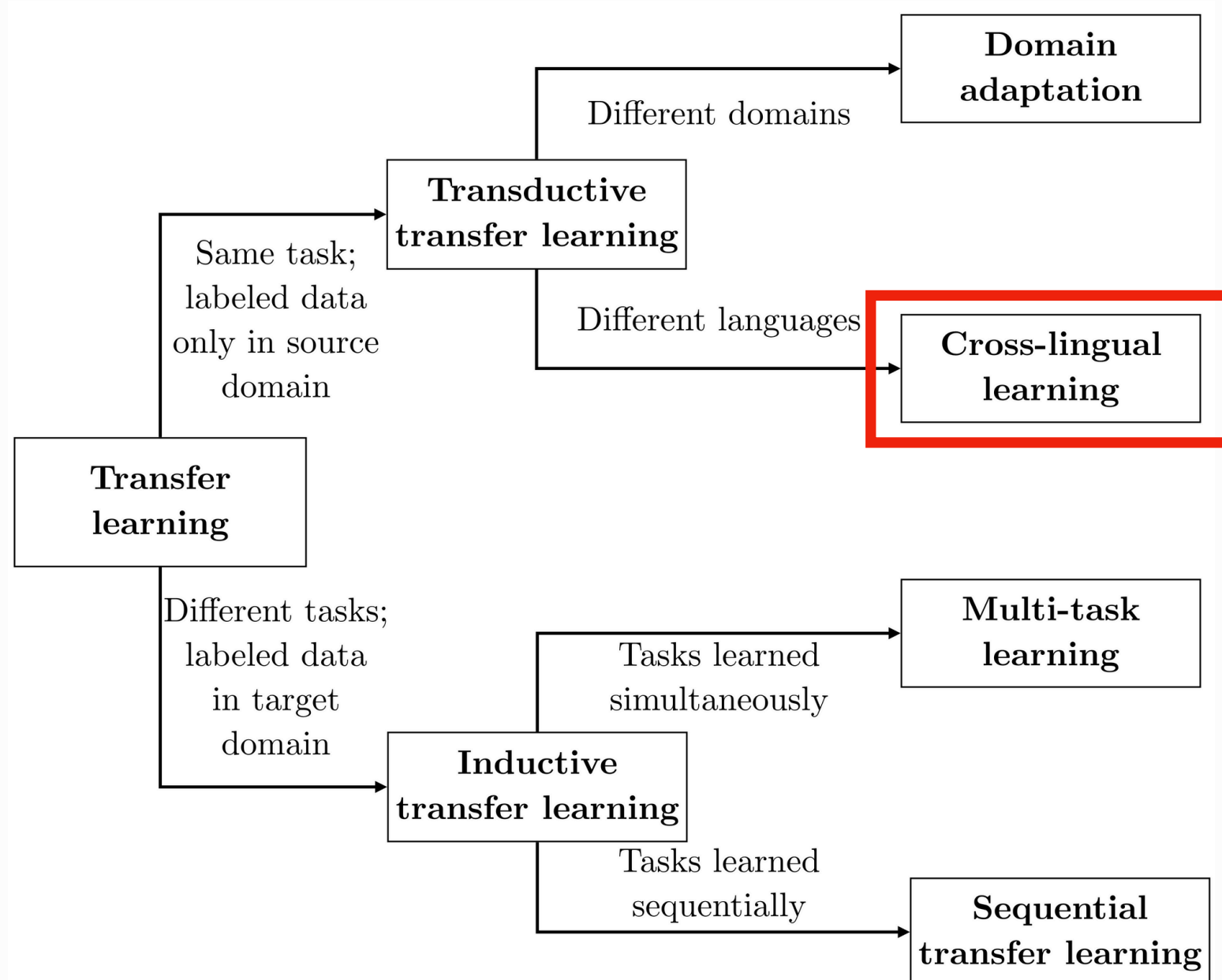
A sentiment analysis model trained on English product reviews (e.g., Amazon) is adapted to analyze sentiment in movie reviews (e.g., IMDb).

The domains are different (products vs. movies), but the task (sentiment classification) remains the same.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Example

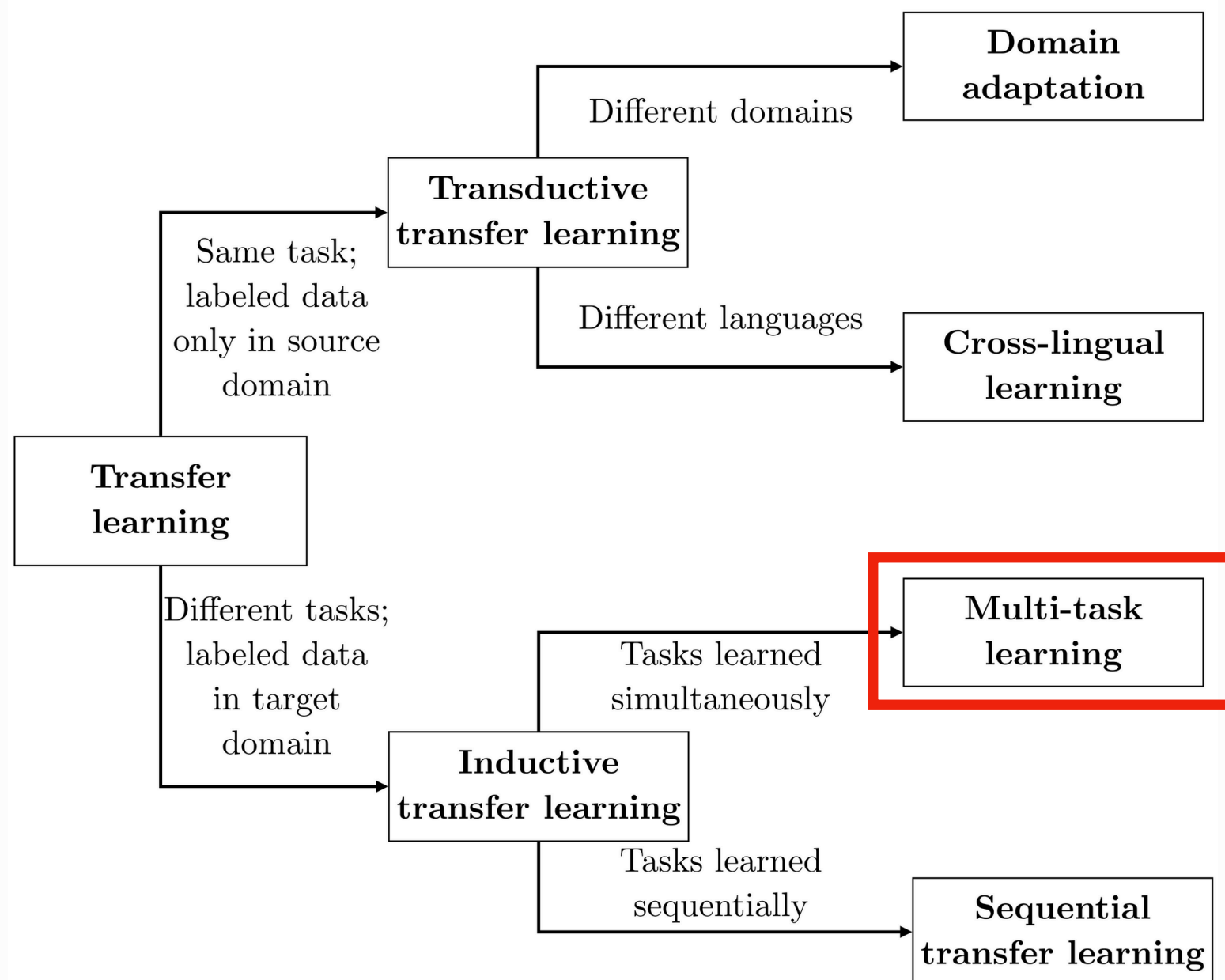
A neural machine translation model trained on English-to-French text is adapted to translate English to Spanish without needing extensive labeled parallel data in English-Spanish pairs.

The domain and task are the same, but the language is different.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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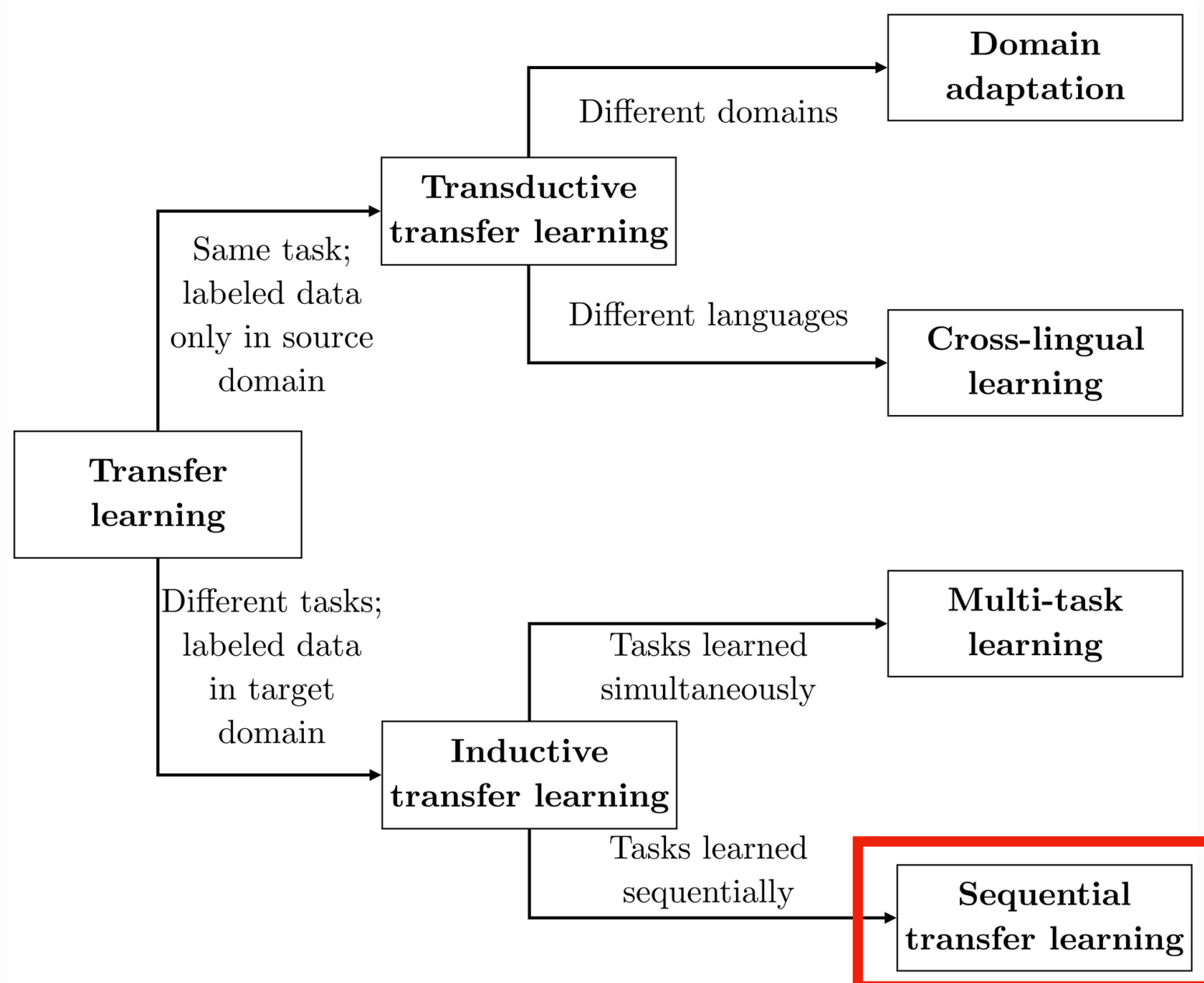
Example

A medical AI model simultaneously learns to classify X-ray images for pneumonia detection and estimate the severity of lung disease. Both tasks share some common knowledge, and learning them together improves overall performance.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Sequential transfer learning is a variant of transfer learning in which a model trained on a source task is subsequently adapted to one or more target tasks in a sequential manner.

It is called "sequential" because the model is progressively adapted in multiple stages rather than being completely retrained from scratch.

Step #1. The model is trained on a large and generic source dataset, which helps it learn fundamental features.

Example. *Medical Image Diagnosis.*



A model (e.g., a convolutional neural network) is first trained on a large, generic dataset such as ImageNet, which contains millions of general images (animals, objects, etc.).

The goal is to allow the model to gain strong **general feature extraction capabilities** without requiring medical images at this stage.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Step #1. The model is trained on a large and generic source dataset, which helps it learn fundamental features.

Step #2. The pre-trained model is further trained on the target dataset or an intermediate dataset, which may be smaller and more specific compared to the source dataset. This process occurs sequentially.

(1) *Freezing Layers.* The early layers of the model (which capture general features) are frozen to preserve the previously acquired knowledge.

(2) *Fine-Tuning.* The final layers are trained with the target data to learn more specific features.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

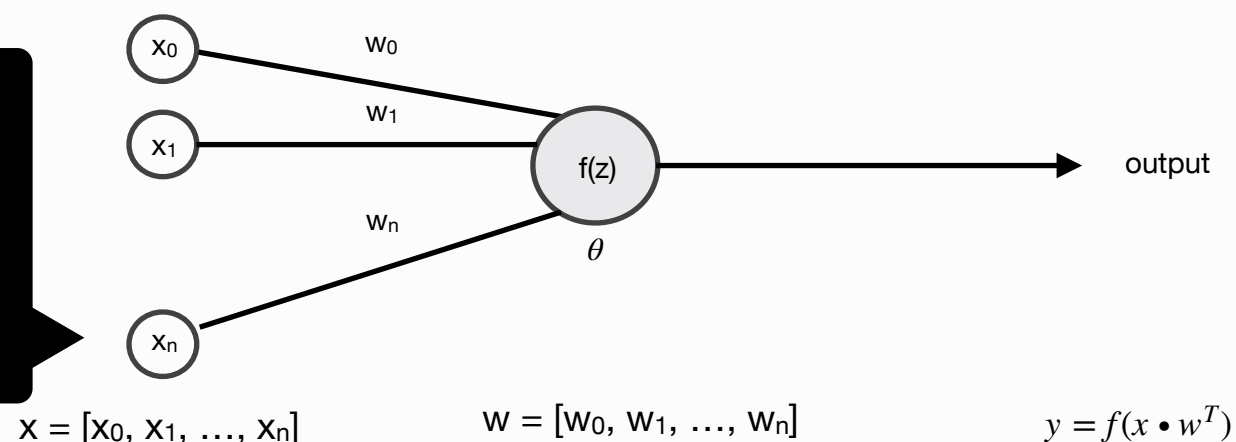
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Example. *Medical Image Diagnosis.*

From a practical perspective, this means "freezing" the weights of the neurons in the early layers while allowing the weights of the later neurons to be modified.



The model can also go through multiple phases of sequential fine-tuning on different datasets, each representing a higher level of specificity.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Example. *Medical Image Diagnosis.*

In medical image diagnosis models, the early layers are responsible for capturing basic visual patterns, such as edges, textures, and general shapes. By freezing these layers, the model retains these fundamental feature representations, preventing unnecessary modifications when adapting to a new medical imaging task.

Very similarly, In language models, the early layers are responsible for capturing basic syntactic patterns, such as sentence structure and word context. By freezing these layers, these general representations are preserved, avoiding unnecessary modifications.

In other terms, the second step aims at refining the knowledge by sequentially add more information, much like we, as humans, do normally!

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Step #3. In the subsequent steps, the target domain data is used to further refine the model. At this stage, only a subset of the model's layers is updated.

These three steps allow us to discuss the basic mechanisms behind LLMs.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

But what type of machine learning models are we talking about? So far, we have only mentioned that they are neural networks...

Transformer: A specific neural network architecture based on the self-attention mechanism, capable of modeling complex relationships between elements in a sequence, regardless of their relative position.

Configuring Artificial Intelligence Solutions: Large Language Models

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaizer@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Łukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

[†]Work performed while at Google Brain.

[‡]Work performed while at Google Research.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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But what type of machine learning models are we talking about? So far, we have only mentioned that they are neural networks...

Transformer: A specific neural network architecture based on the self-attention mechanism, capable of modeling complex relationships between elements in a sequence, regardless of their relative position.

As a matter of fact, this mechanism represents one of the most significant innovations of the century. Traditionally, neural networks process a sequence *element by element*, making it more challenging to capture relationships between distant words in the sequence.

Transformers, instead, use the self-attention mechanism to analyze each element of the sequence in *relation to all others simultaneously*, assigning a weight that represents their relative importance.

This makes them particularly well-suited for natural language processing tasks and beyond, such as machine translation, text completion, and text generation.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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What does "attention" mean?

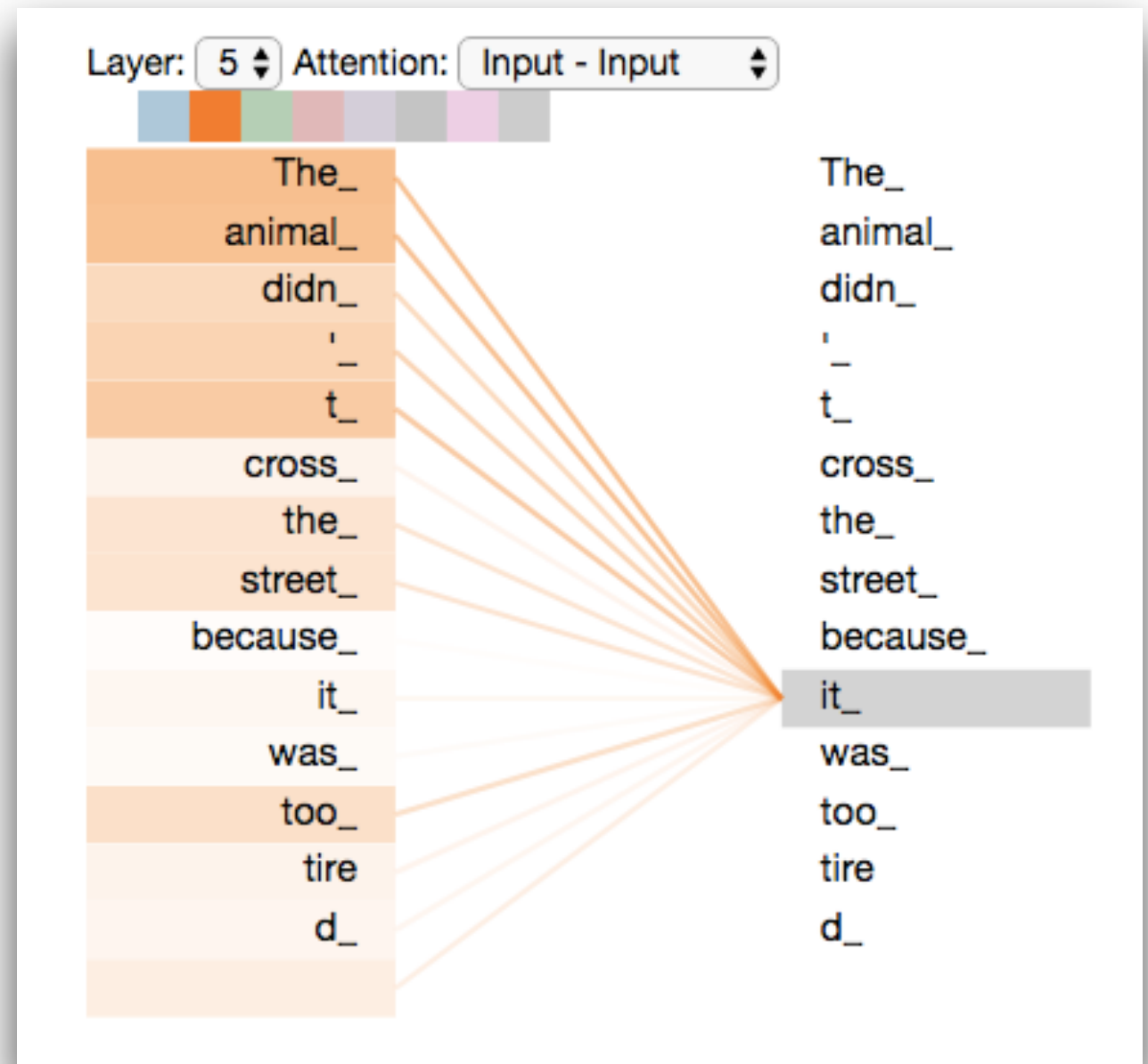
Given the sentence:

*"The animal didn't cross the street because **it** was too tired".*

The mechanism, by considering the context of the entire sentence, is able to assign a higher probability that "**it**" refers to the animal rather than the street.

In traditional models, this would not be possible, as each word would be considered individually.

Let's elaborate on this matter...

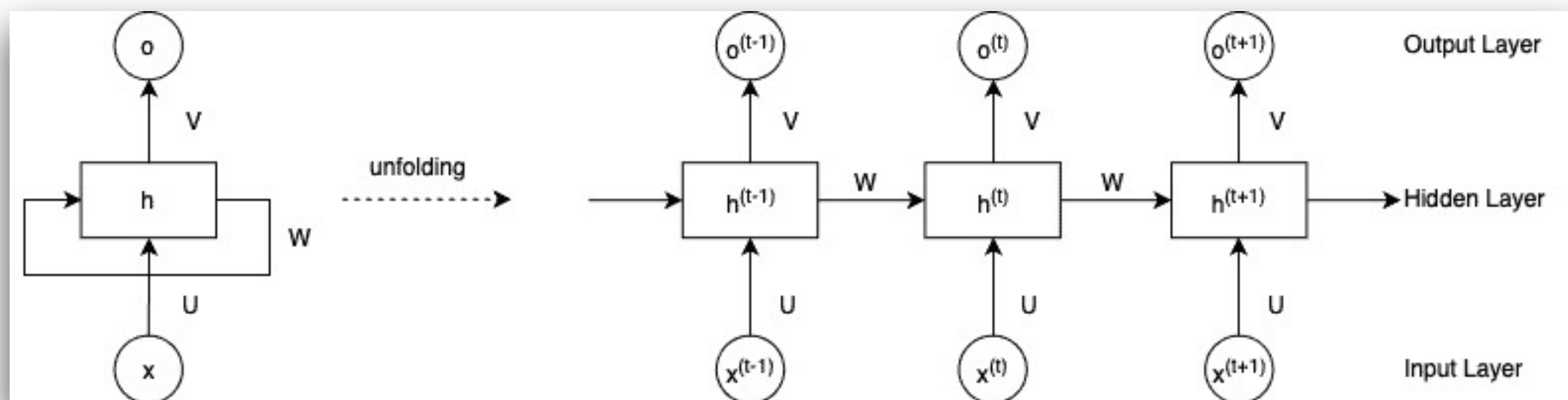


Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

This is the basic structure of a Recurrent Neural Network:



Where x , h , o are the input sequence, hidden state, and the output sequence, respectively. U , V , and W are the training weights.

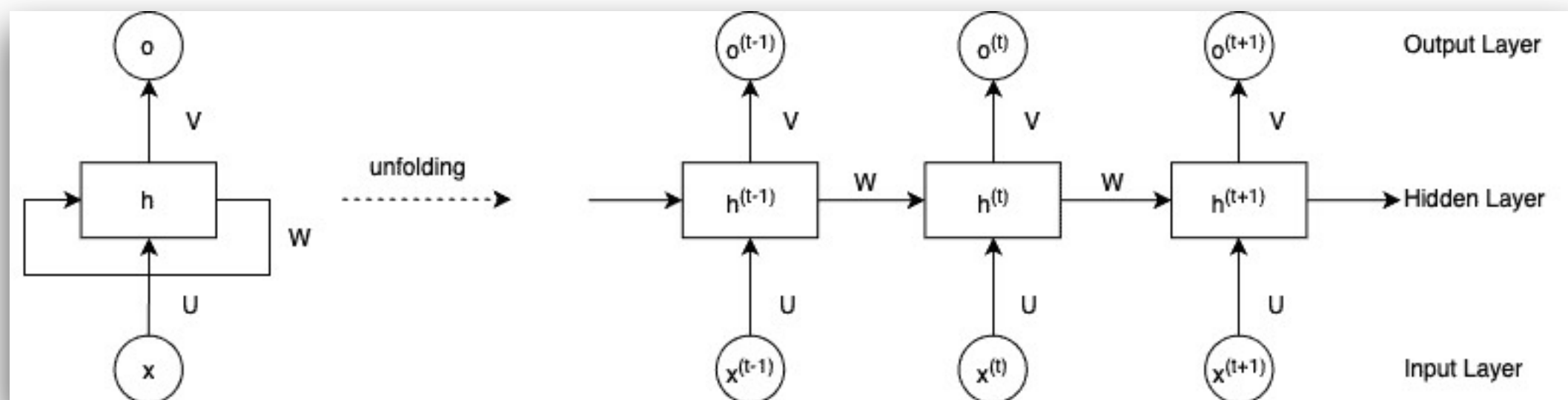
RNNs are sequential models that process data *one element at a time*, maintaining an internal hidden state that is updated at each step. They operate in a **recurrent** manner, where the output at each step depends on the previous hidden state and the current input.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

This is the basic structure of a Recurrent Neural Network:



For instance, if applied to text generation, RNNs would process text one word (or character) at a time, using previous words to predict the next one.

The model's hidden state carries memory from previous words, but as the sentence grows longer, earlier words get "forgotten" (**vanishing gradient** problem).

This means that RNNs only remember *recent words* well, lacking long-range coherence.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

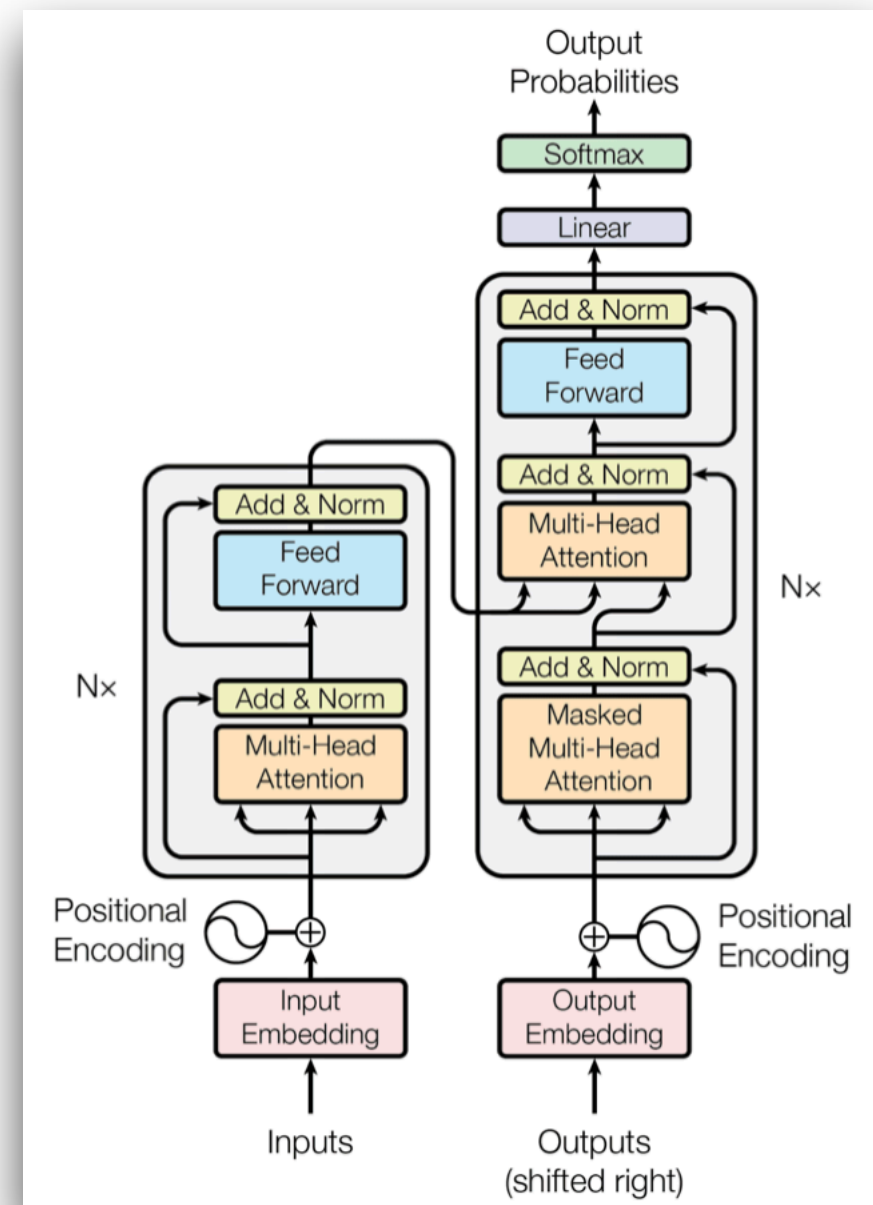
Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Transformers, instead, use a way more complex architecture based on two main components:

- (1) *Encoder*. A module that takes an input sequence and transforms it into a representation understandable by the model.
- (2) *Decoder*. A module that generates an output sequence based on the representation created by the encoder and the previously generated elements.

The architecture is composed of eight main steps that allow the model to decompose the problem and refine the solution.

Let's try to better understand how this architecture actually works...



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Step #1. *Input Embedding & Positional Encoding*

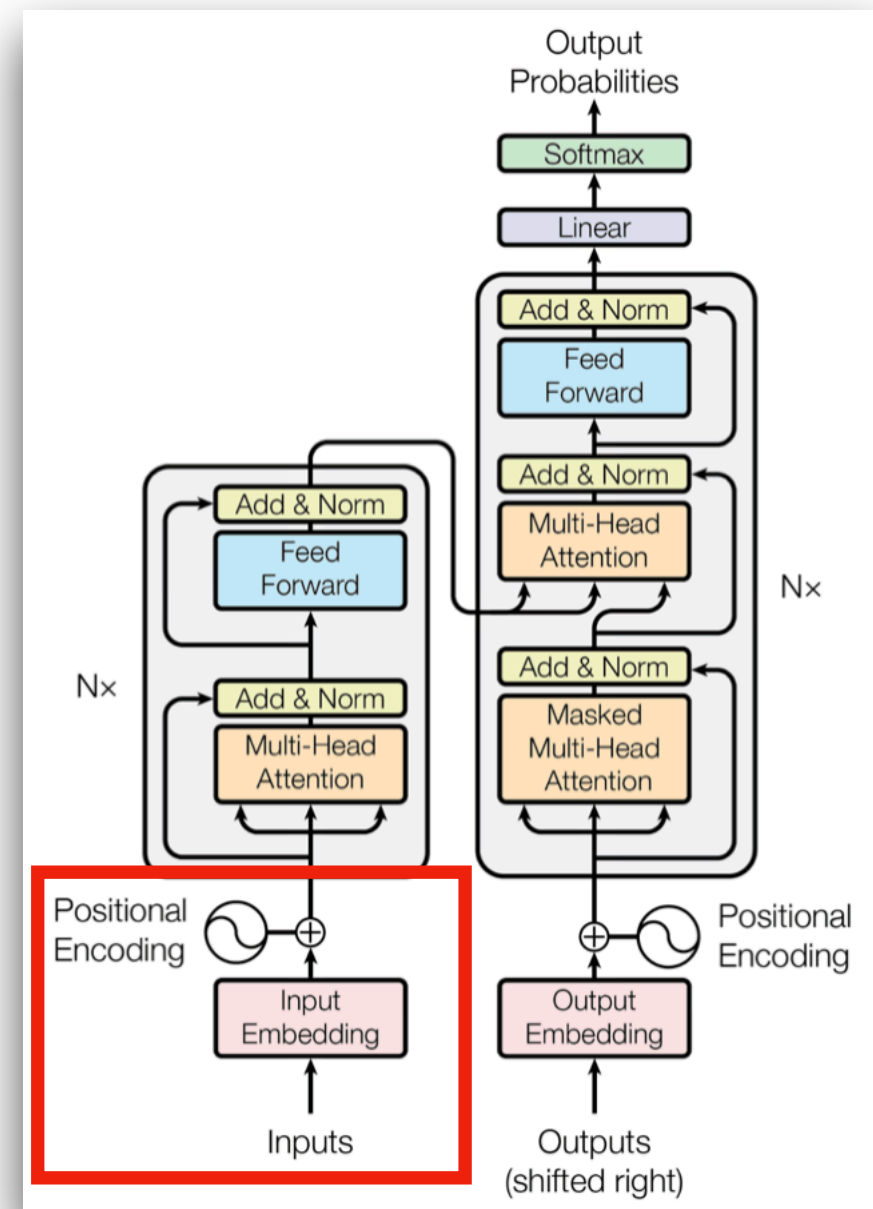
The input text is converted into numbers (embeddings) - machines do not understand words. This provides an initial weight of the word, depending on the embedding employed, e.g., in Word2Vec certain words may be considered inherently more important than others.

Since Transformers do not process words in order like RNNs, positional encoding is added to *indicate the position of each word in the sentence*.

Analogy



Imagine writing words on index cards, but to know their order, you also write a number on each card.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Step #2. Multi-Head Attention

This mechanism analyzes relationships between words in the input sentence. Each word is compared to every other word simultaneously to understand their importance.

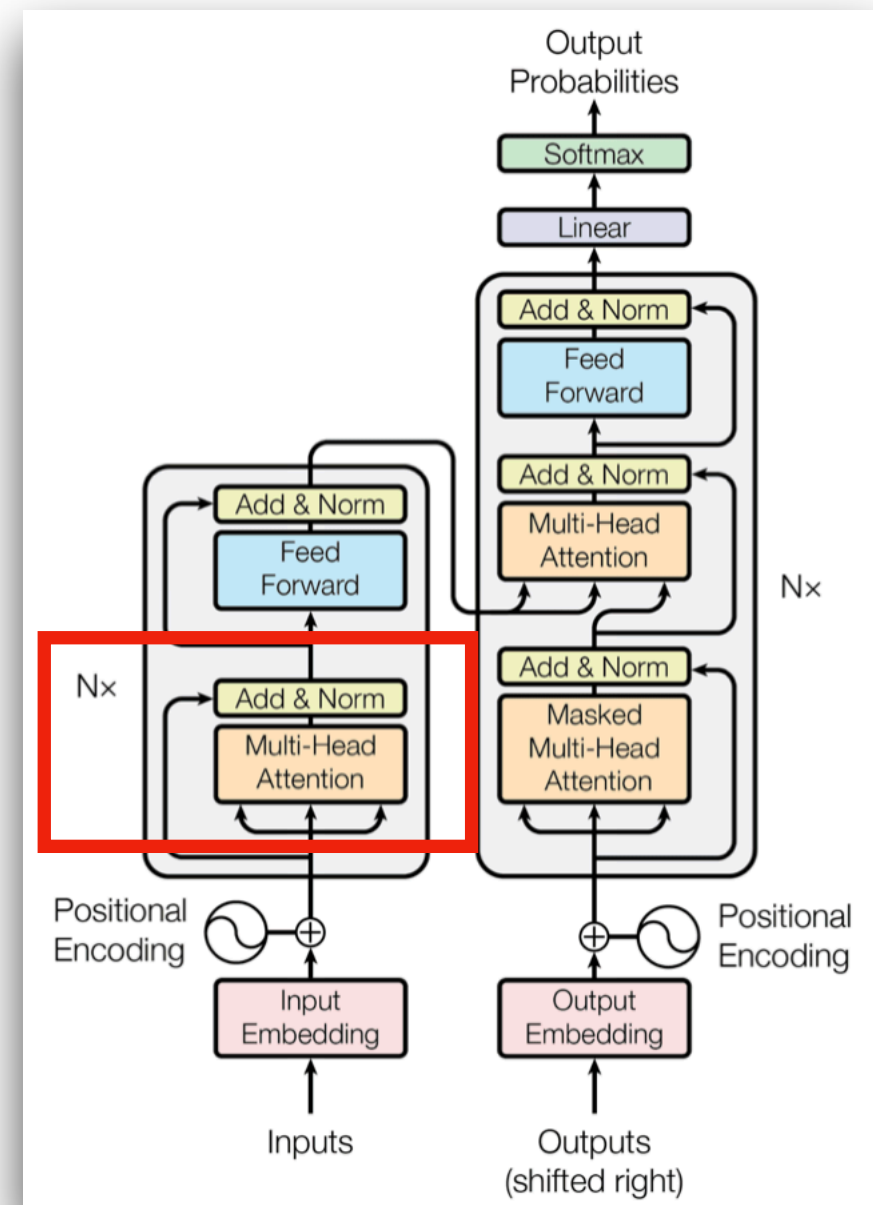
Each word in a sentence is converted into three vectors using learned weights:

Query (Q): Represents the numerical representation provided by its embedding.

Key (K): Each word embedding is multiplied by an (initially random) learned weight matrix to create K - the weight will be refined during training.

Value (V): Holds the actual value of the word.

In other terms, the key vector attempts to weigh the general importance of a word (assigned by the embeddings) based on the specific importance of the word in the given context (the sentence).



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Step #2. *Multi-Head Attention*

Word	Query (Q)	Key (K)	Value (V)
The	[0.1]	[0.2]	"The"
Cat	[0.5]	[0.8]	"Cat"
Sat	[0.3]	[0.4]	"Sat"
On	[0.2]	[0.3]	"On"
The	[0.1]	[0.2]	"The"
Mat	[0.4]	[0.7]	"Mat"
Because	[0.2]	[0.3]	"Because"
It	[0.9]	[0.5]	"It"
Was	[0.3]	[0.4]	"Was"
Tired	[0.6]	[0.9]	"Tired"

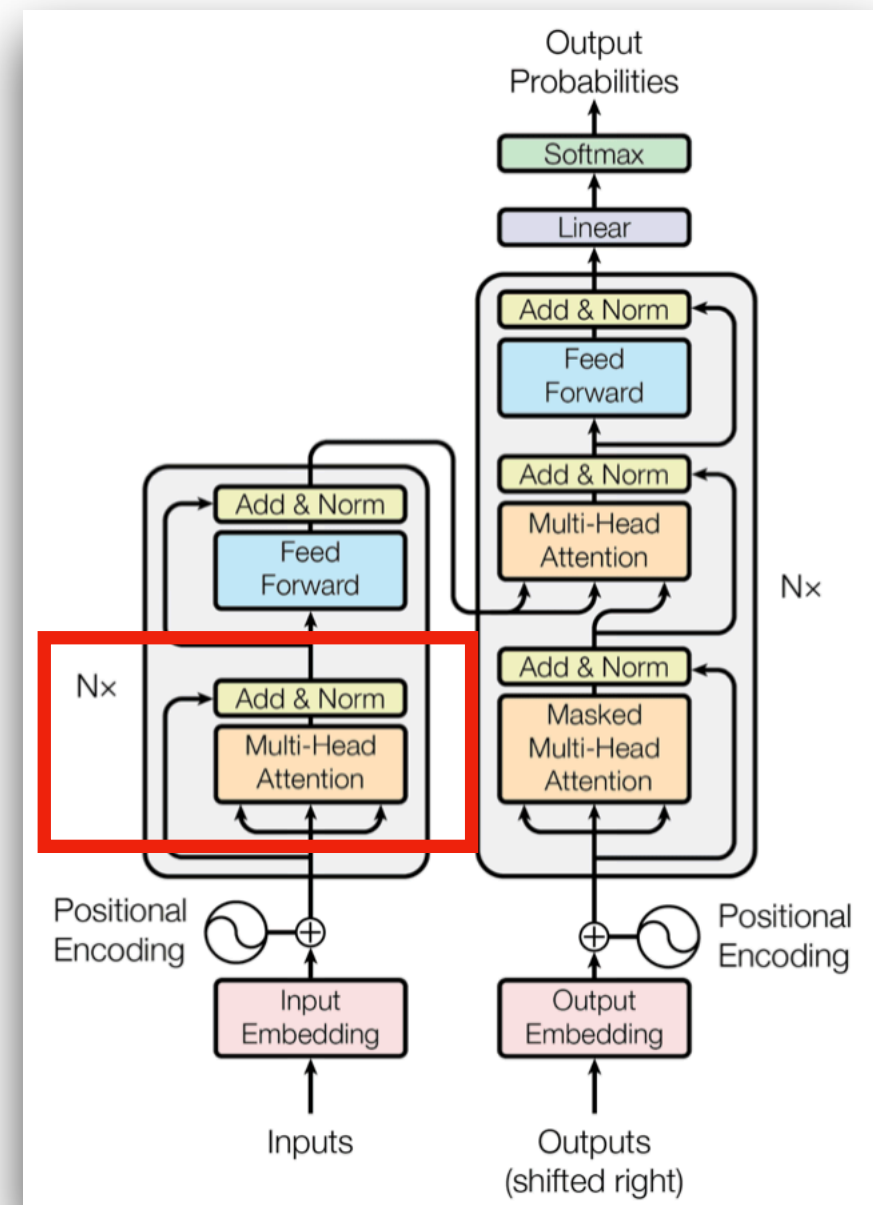
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Step #2. *Multi-Head Attention*

Afterwards, the *attention* score is computed by applying a dot product (i.e., multiplication) between a query and each key value within the considered sentence.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Step #2. *Multi-Head Attention*

Word	Key (K)	Dot Product with "it"'s Query ($0.9 \times \text{Key}$)	Attention Score
The	[0.2]	0.18	Low
Cat	[0.8]	0.72	High ✓
Sat	[0.4]	0.36	Medium
On	[0.3]	0.27	Low
The	[0.2]	0.18	Low
Mat	[0.7]	0.63	Medium
Because	[0.3]	0.27	Low
It	[0.5]	0.45	Medium
Was	[0.4]	0.36	Medium
Tired	[0.9]	0.81	High ✓

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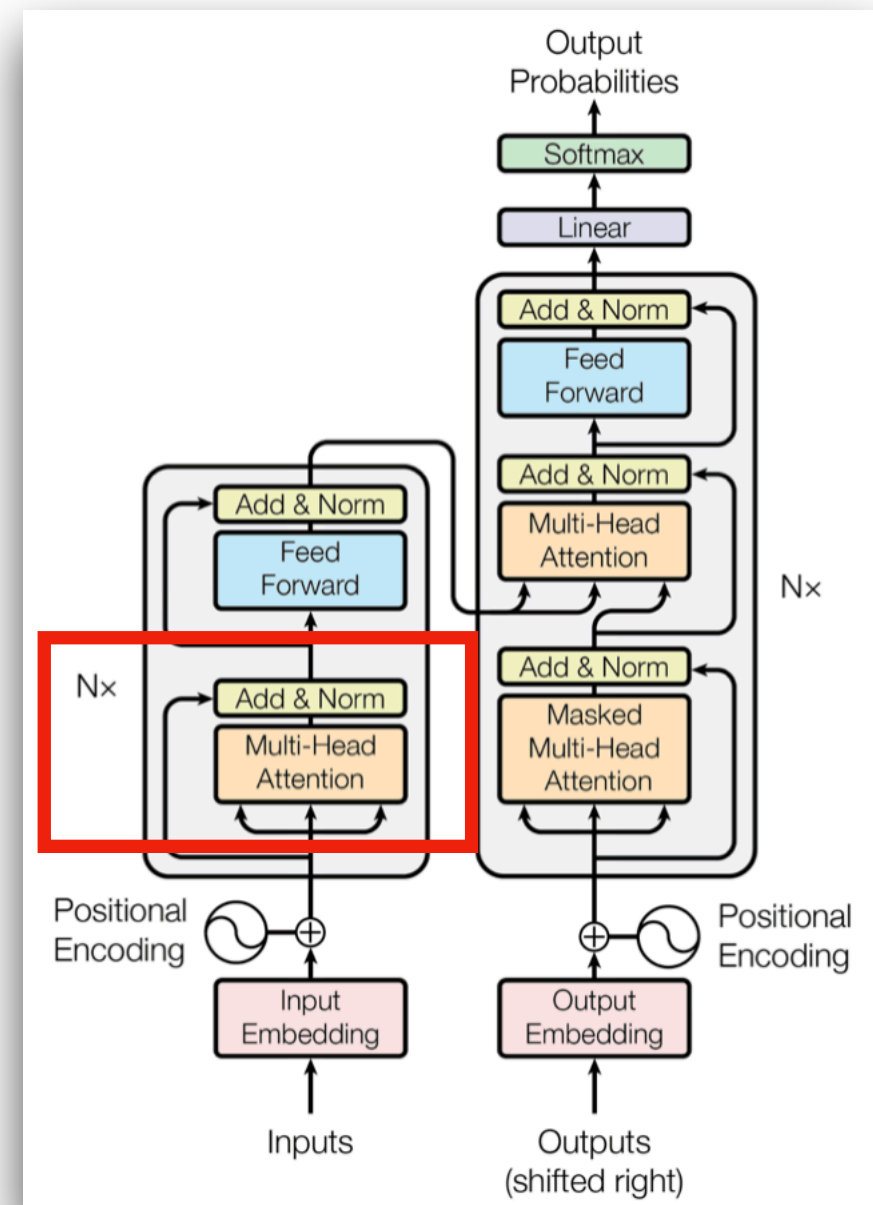
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Afterwards, the *attention* score is computed by applying a dot product (i.e., multiplication) between a query and each key value within the considered sentence.

The raw attention scores are normalized using a Softmax function to convert them into probabilities.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Step #2. Multi-Head Attention

Word	Raw Score	Softmax Computation	Final Attention Weight
The	0.18	$e^{0.18} / (e^{0.18} + e^{0.72} + e^{0.36} + e^{0.27} + e^{0.18} + e^{0.63} + e^{0.27} + e^{0.45} + e^{0.36} + e^{0.81})$	0.077
Cat	0.72 ✓	$e^{0.72} / \text{sum}$	0.131 ✓
Sat	0.36	$e^{0.36} / \text{sum}$	0.092
On	0.27	$e^{0.27} / \text{sum}$	0.084
The	0.18	$e^{0.18} / \text{sum}$	0.077
Mat	0.63	$e^{0.63} / \text{sum}$	0.120
Because	0.27	$e^{0.27} / \text{sum}$	0.084
It	0.45	$e^{0.45} / \text{sum}$	0.101
Was	0.36	$e^{0.36} / \text{sum}$	0.092
Tired	0.81 ✓	$e^{0.81} / \text{sum}$	0.144 ✓

This starts to explain why, in some cases, these models hallucinate!

The model leans more toward associating "it" with "tired" in this specific case. This behavior can change depending on the dataset and training process

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

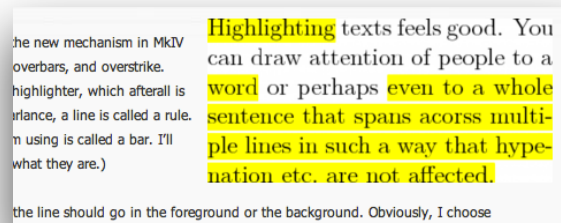
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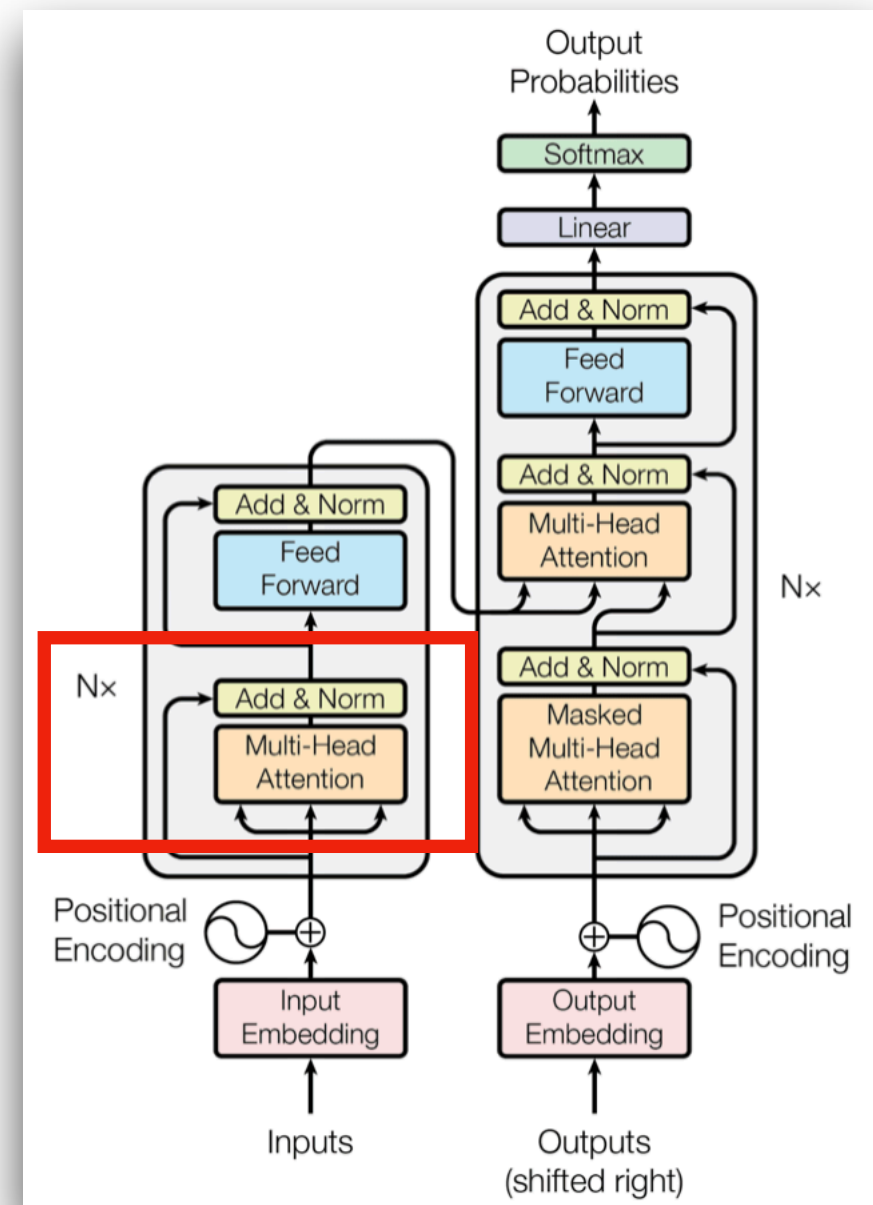
The raw attention scores are normalized using a Softmax function to convert them into probabilities.

Analogy



the new mechanism in MkIV overbars, and overstrike. highlighter, which afterall is arlance, a line is called a rule. n using is called a bar. I'll what they are.)
the line should go in the foreground or the background. Obviously, I choose

It's like reading a paragraph and underlining words that seem important in relation to each other.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Step #3. *Feed-Forward & Normalization*

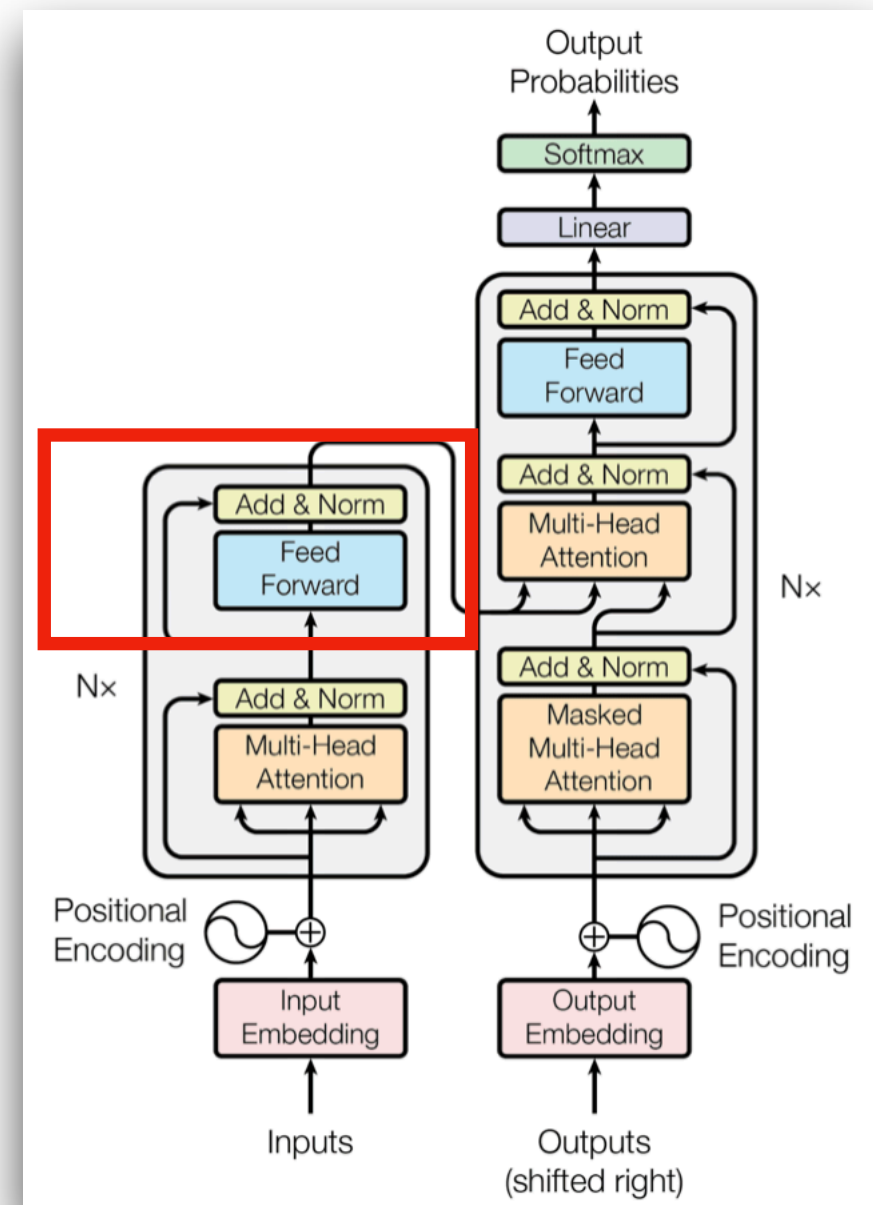
A feed-forward neural network (i.e., a network where information flows in one direction) processes the attention output to refine the information.

Normalization helps stabilize training and improve efficiency. This step is repeated multiple times (Nx) to improve learning.



Analogy

Imagine summarizing a book chapter multiple times, refining it with each revision.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

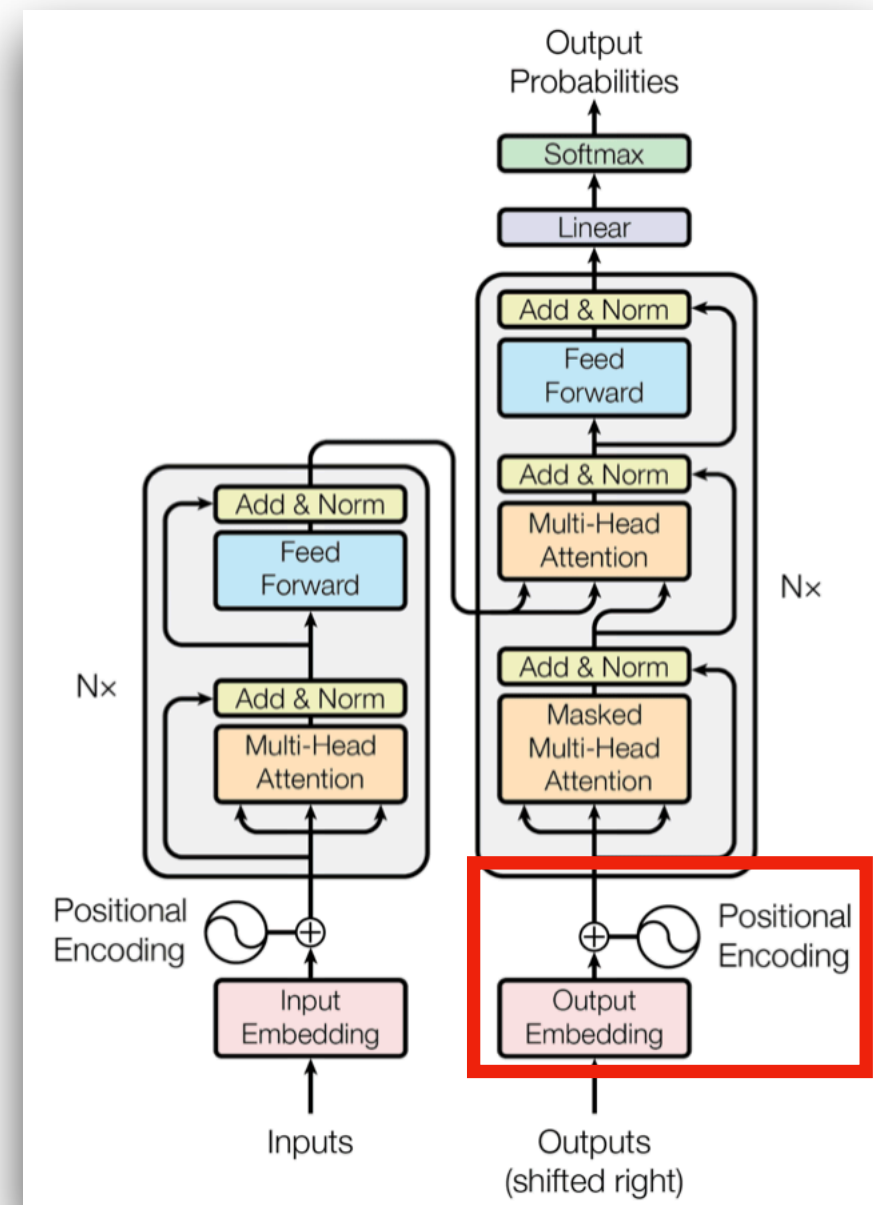
Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Step #4. *Input Embedding & Positional Encoding*

This step depends on the task we are using the architecture for. As an example, in *machine translation*, the decoder receives the target language sentence shifted to the right - this ensures the model generates the next word step by step rather than seeing the whole answer at once. In *text generation*, the model does not have a target sentence upfront, but gets a prompt.

In any case, the process is similar to what discussed for the encoder: words are converted in a numerical form using an embedding and then marked based on their position in the text.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

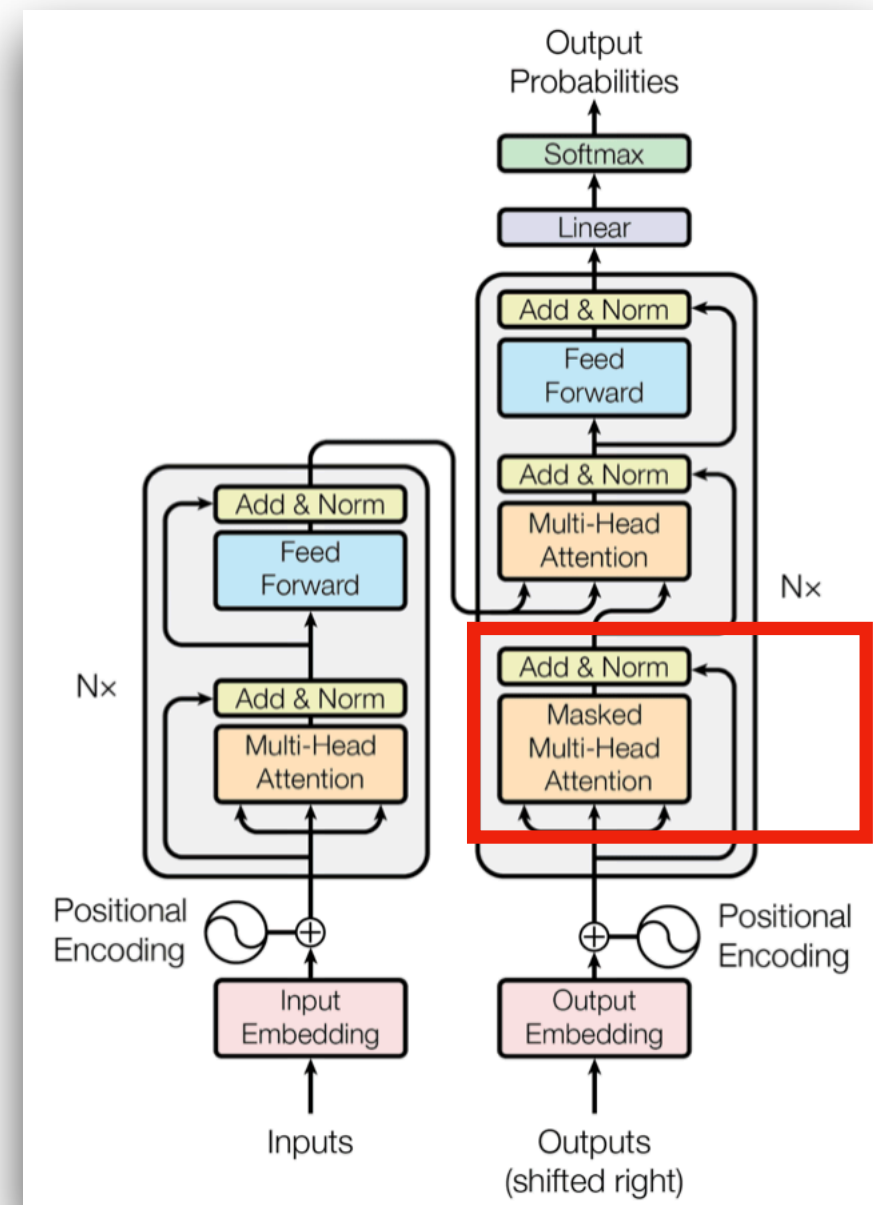
Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Step #5. *Masked Multi-Head Attention*

Unlike the encoder, which processes all words simultaneously, the decoder must generate an output sequence one word at a time. If the decoder could "see" future words while predicting the next one, it would cheat by using information it should not have yet.

To prevent this, masked self-attention is applied. This mechanism blocks the model from attending to words it has not yet generated, ensuring that each word is predicted based only on the words that came before it.

The goal of multi-head attention is to compute attention scores by considering how relevant each previously generated word is to the next word being predicted. This representation is then passed to the final layers, which convert it into a probability distribution over the vocabulary, ultimately selecting the most likely next word.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

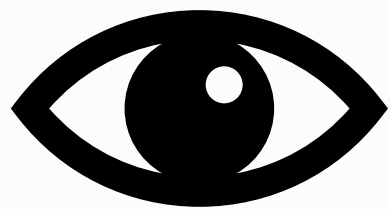
Configuring Artificial Intelligence Solutions: Large Language Models

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Step #6. *Multi-Head Attention with Encoder*

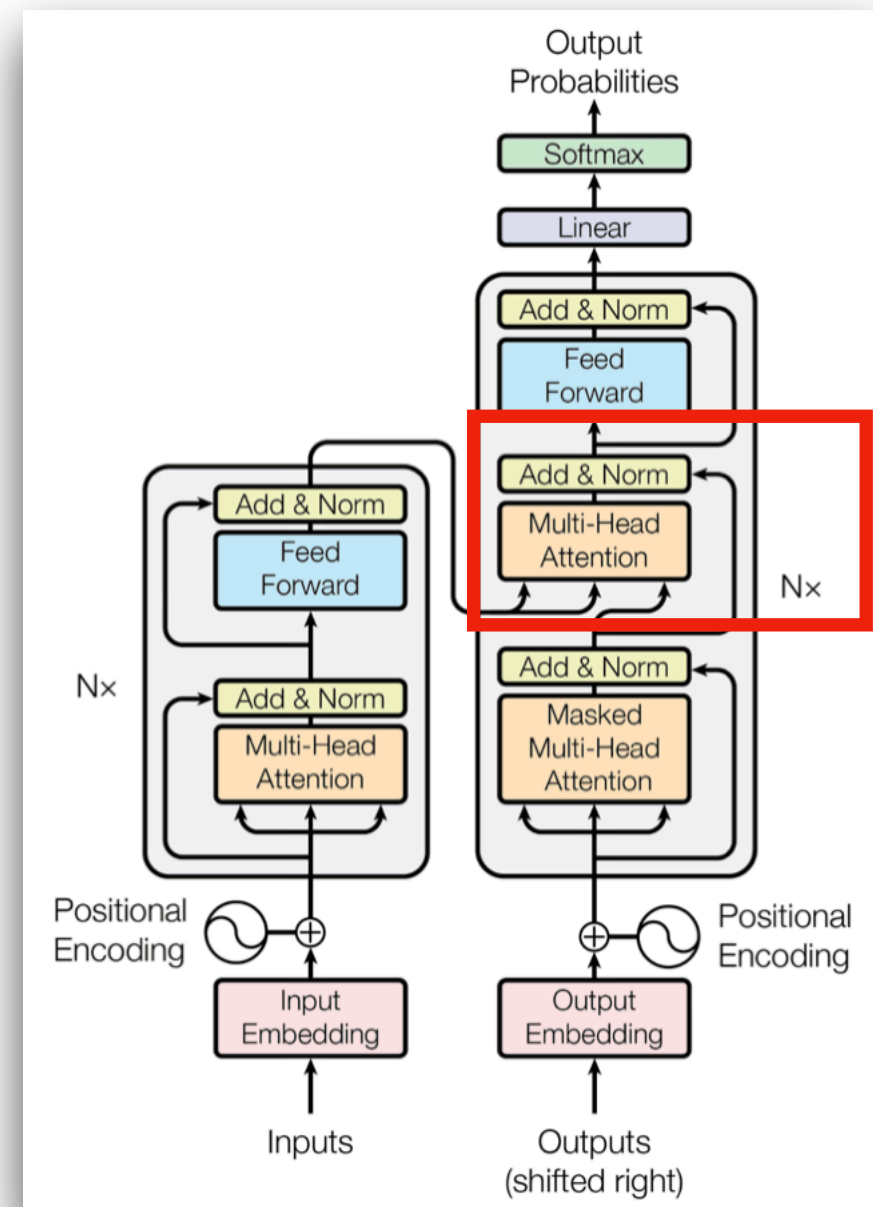
Shall we stop at Step #5, the decoder would generate words purely based on itself without knowing the meaning of the original input. This would not make sense in cases like *language translation*, where we have a precise input sequence to translate. This step allows the model to align each generated word with relevant words from the original input sentence and ensure that the text generated faithfully represent the input text.

Language generation does **not** include this step, as the task involves generating text freely.



Analogy

If someone explains something in English and you need to translate it into Italian, you keep checking back to their words for accuracy



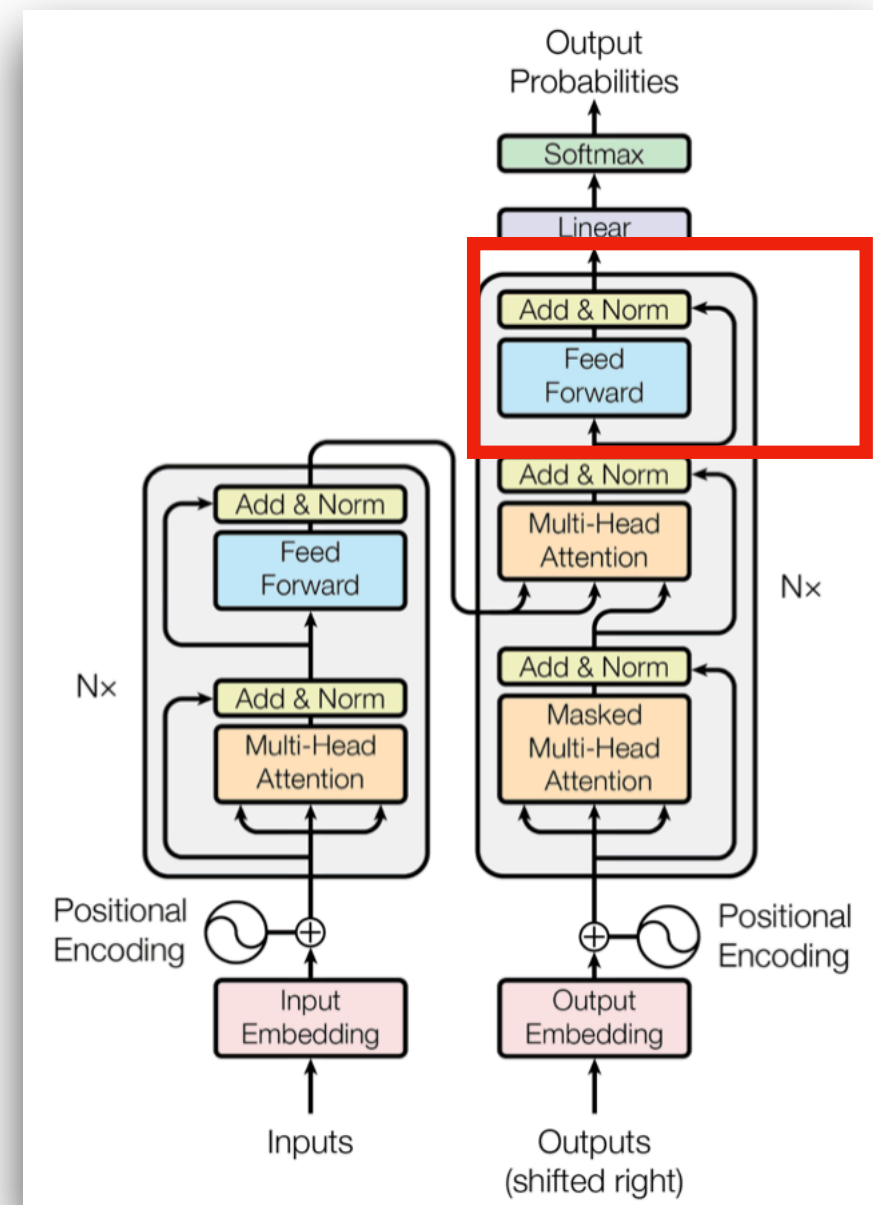
Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Step #7. *Feed-Forward & Normalization*

Like in the encoder, this step refines the generated sentence to improve fluency and accuracy. This step is repeated multiple times (Nx) to improve learning.



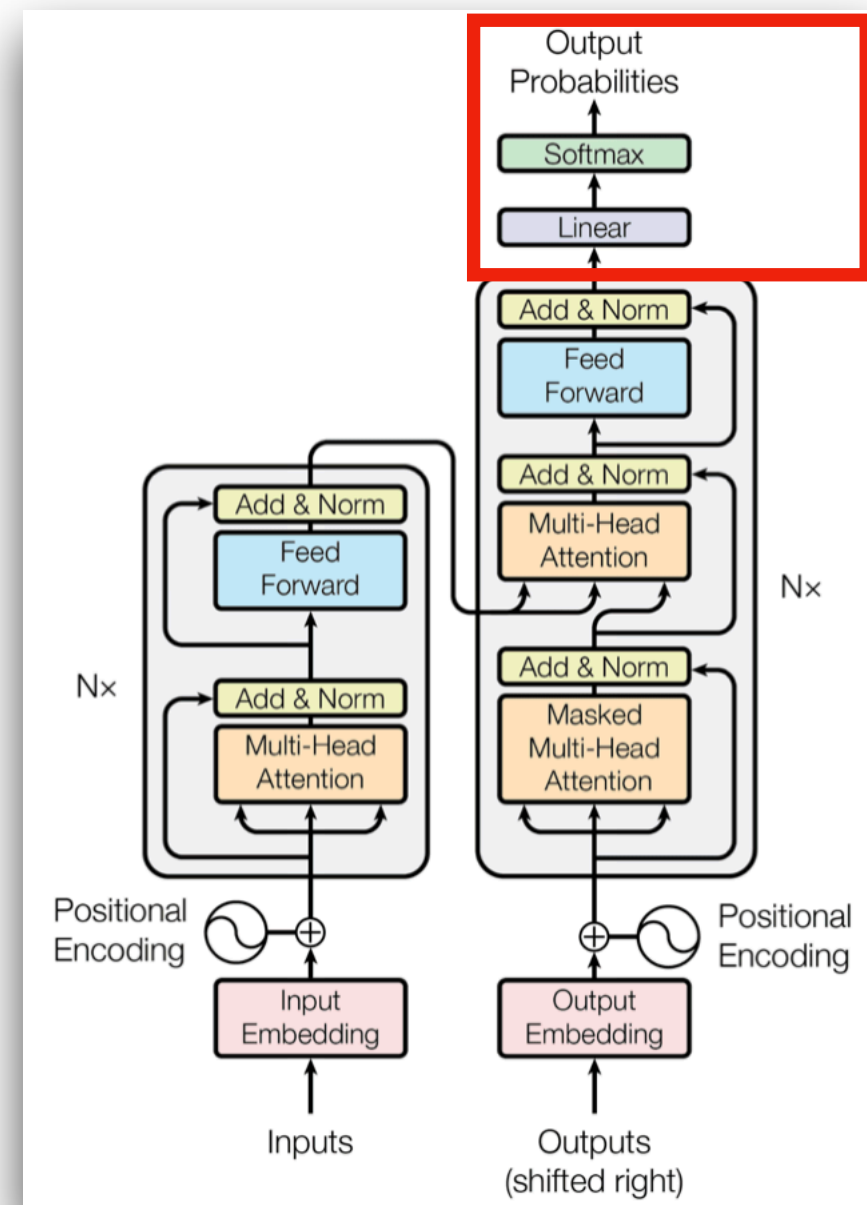
Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Step #8. *Linear Layer & Softmax*

This step transforms the attention scores into a probability distribution over the vocabulary. This is again done by using a Softmax function - the difference is that the final attention weights are not computed on the sentence itself (as done in the Encoder) but on all words in a vocabulary.



Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Step #8. *Linear Layer & Softmax*

How many words are available in a vocabulary?

Model	Vocabulary Size
GPT-2	~50,000 tokens
GPT-3	~50,257 tokens
BERT	~30,522 tokens
T5	~32,000 tokens
GPT-4	100,256 tokens
GPT-4o	199,997 tokens

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Step #8. *Linear Layer & Softmax*

How many words are available in a vocabulary?

Starting from the probabilities computed, how can a model select the next word to generate?

Strategy	How It Works	Pros	Cons
Greedy Search	Pick the highest probability word.	Fast & simple.	Can lead to repetitive or boring text.
Beam Search	Keeps multiple candidate sequences and picks the best.	Produces better quality text.	More computationally expensive.
Top-k Sampling	Picks from the top k most likely words randomly.	Adds diversity to generation.	Can introduce randomness.
Temperature Scaling	Adjusts Softmax probabilities to control randomness.	Higher temp = more creative text.	Too high temp = nonsense text.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

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Based on this architecture, several well-known models have been developed:

(1) *BERT (Bidirectional Encoder Representations from Transformers)*

A pre-trained model designed solely as an **encoder**. BERT analyzes each word in relation to all other words in the sequence, both preceding and following, to **capture the most comprehensive meaning possible**.

(2) *T5 (Text-to-Text Transfer Transformer)*

A pre-trained model designed as both an **encoder** and a **decoder**. It handles a wide range of language tasks by treating them all as **text-to-text conversion problems**. This approach requires an **encoder** to understand the input and a **decoder** to generate the desired output, e.g., translation.

(3) *GPT (Generative Pre-trained Transformer)*

A pre-trained model designed solely as a **decoder**. Its primary goal is to **generate text sequentially**, relying on what has been produced so far.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

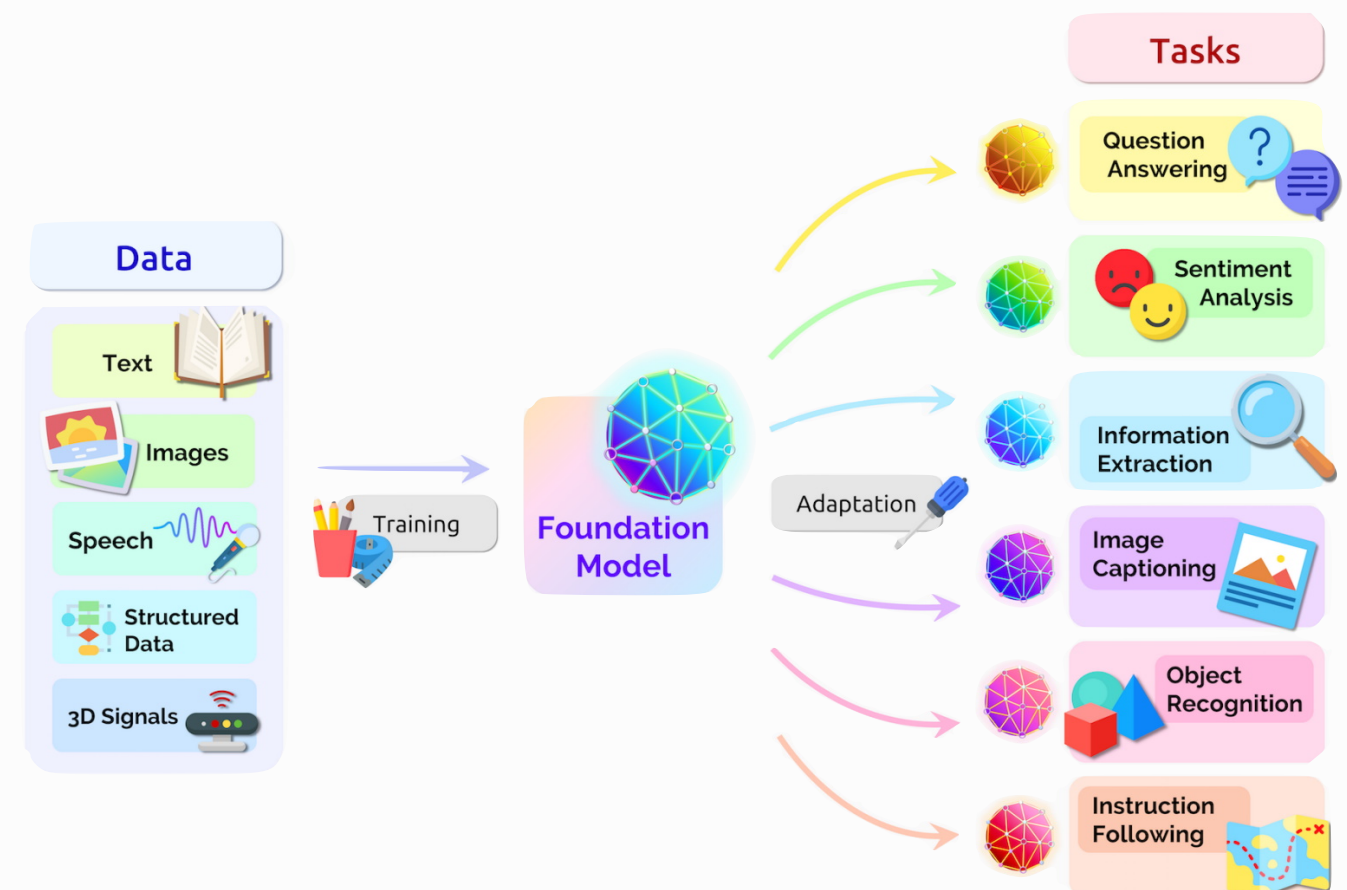
Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Here we are: Large Language Models. These are just one of many so-called "*foundational models*", which are pre-trained on vast amounts of unsupervised data and capable of generalizing to multiple tasks through fine-tuning or prompting.

Large Language Model: GPT3.5 ,
LLAMA, PaLM, etc.

Computer Vision Foundation Model: DALL-E, Generative Adversarial Networks (GAN), Variational Auto-Encoder (VAE)

Multimodal Foundation Model: Contrastive Language-Image Pre-Training (CLIP), GPT-4, ViLBERT



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

LLMs are trained on multiple Natural Language Processing tasks, including:

Language understanding

Sentiment Analysis: Determining the sentiment of a text, that is, identifying whether the text expresses a positive, negative, or neutral sentiment.

Language Detection: Identifying the language of a given input text.

Part-of-Speech Tagging: Identifying parts of speech in a text, such as nouns, verbs, adjectives, etc.

Language generation

Question-Answering: Providing answers to questions based on the information available in a given text or a knowledge base.

Summarization: Generating a summary of a longer text.

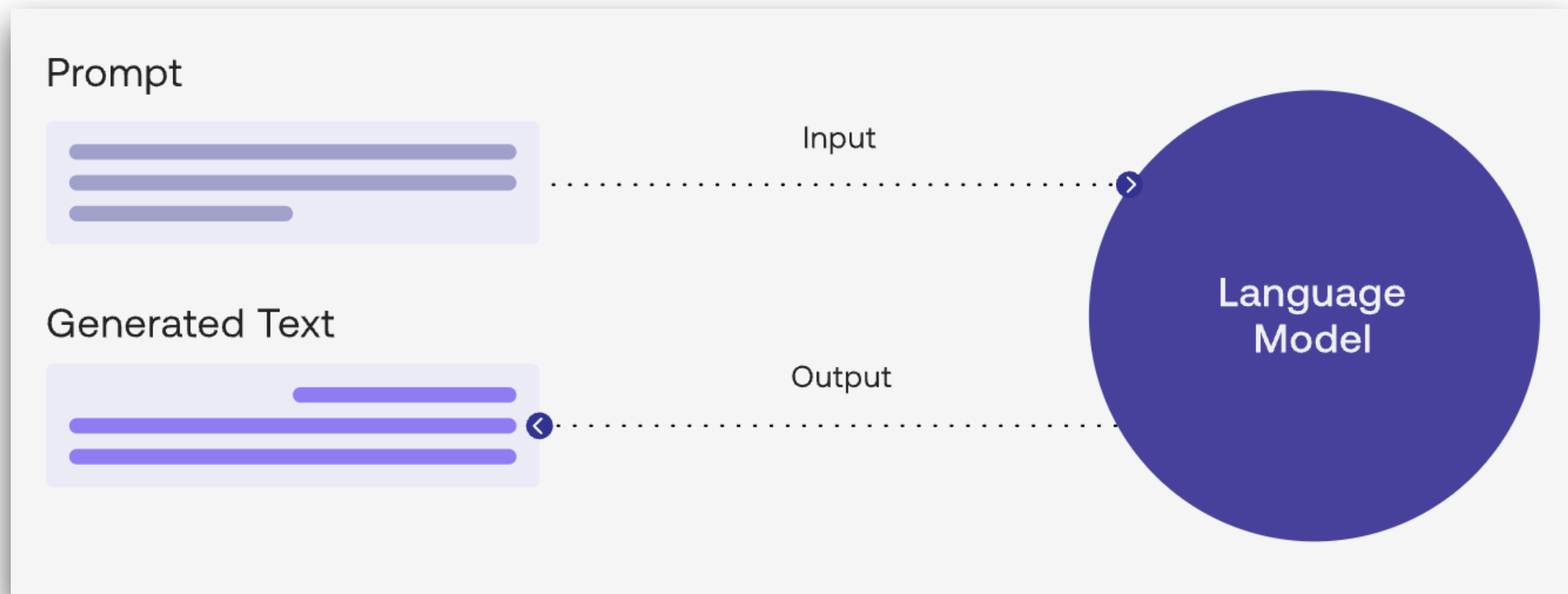
Code Generation: Generating code snippets from a natural language description of the desired functionality.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

LLMs have gained popularity due to the **interaction mechanism** that allows users to engage with them, e.g., the Chat interface of ChatGPT.

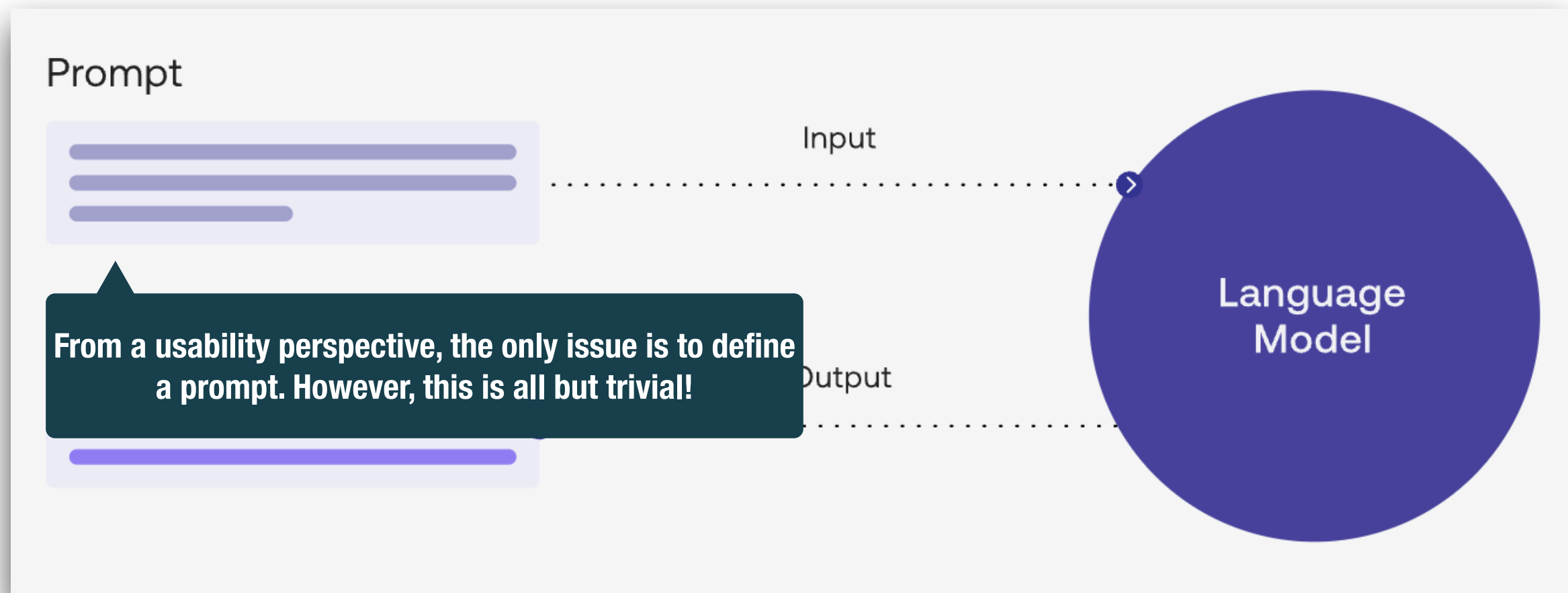


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Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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There are, indeed, various challenges with the prompt definition:

Ambiguity: Creating **clear and specific prompts** is essential. **Ambiguous prompts** can lead to **misleading or irrelevant** responses.

Context Sensitivity: Prompts often require an **understanding of context** that the model may lack or misinterpret.

Model Limitations: LLMs are restricted to the **knowledge available in their training data**, which may lead to **outdated or suboptimal** responses.

Dynamic Knowledge: Even well-designed prompts can only yield responses within the **limits of the model's capabilities** and **knowledge base**.

Complexity: There is a balance between providing **enough details** and making prompts **overly complex** or **lengthy**, which could confuse the model.

All of this indicates the need to develop **well-structured strategies** for querying LLMs—this necessity forms the foundation of **prompt engineering**.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Suboptimal prompts may have further consequences:

Error Proneness: LLMs make mistakes frequently!

Is Stack Overflow Obsolete? An Empirical Study of the Characteristics of ChatGPT Answers to Stack Overflow Questions

Samia Kabir
Purdue University
West Lafayette, USA
kabirs@purdue.edu

Bonan Kou
Purdue University
West Lafayette, USA
koub@purdue.edu

David N. Udo-Imeh
Purdue University
West Lafayette, USA
dudoimeh@purdue.edu

Tianyi Zhang
Purdue University
West Lafayette, USA
tianyi@purdue.edu

ABSTRACT

Q&A platforms have been crucial for the online help-seeking behavior of programmers. However, the recent popularity of ChatGPT is altering this trend. Despite this popularity, no comprehensive study has been conducted to evaluate the characteristics of ChatGPT's answers to programming questions. To bridge the gap, we conducted the first in-depth analysis of ChatGPT answers to 517 programming questions on Stack Overflow and examined the correctness, consistency, comprehensiveness, and conciseness of ChatGPT answers. Furthermore, we conducted a large-scale linguistic analysis, as well as a user study, to understand the characteristics of ChatGPT answers from linguistic and human aspects. Our analysis shows that 52% of ChatGPT answers contain incorrect information and 77% are verbose. Nonetheless, our user study participants still preferred ChatGPT answers 35% of the time due to their comprehensiveness and well-articulated language style. However, they also overlooked the misinformation in the ChatGPT answers 39% of the time. This implies the need to counter misinformation in ChatGPT answers to programming questions and raise awareness of the risks associated with seemingly correct answers.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI;
• Software and its engineering; • General and reference → Empirical studies;

KEYWORDS

stack overflow, q&a, large language model, chatgpt, misinformation

Conference on Human Factors in Computing Systems (CHI '24), May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 17 pages. <https://doi.org/10.1145/3613904.3642596>

1 INTRODUCTION

Programmers often resort to online resources for a variety of programming tasks, e.g., API learning, bug fixing, comprehension of code or concepts, etc. [70, 75, 86]. A vast majority of these help-seeking activities include frequent engagement with community Q&A platforms such as Stack Overflow (SO) [69, 70, 84, 86]. The emergence of *Large Language Models (LLMs)* has demonstrated the potential to transform the online help-seeking patterns of programmers. In November 2022, ChatGPT [61] was released and quickly gained significant attention and popularity among programmers. There have been increasing debates about whether and when ChatGPT would replace prominent search engines and Q&A forums among researchers and industrial practitioners [22, 68].

Despite the rising popularity of ChatGPT, there are also many increasing concerns. Previous studies show that LLMs can acquire factually incorrect knowledge during training and propagate the incorrect knowledge to generated content [9, 33, 35, 39, 56]. Besides, LLMs often generate fabricated texts that mimic truthful information and are hard to recognize, especially for users who lack the expertise [14, 21, 29]. Like other LLMs, ChatGPT is also plagued with these issues [15, 41, 50, 58]. The prevalence of misinformation, which can easily mislead users, has prompted Stack Overflow to impose a ban on answers generated by ChatGPT [64].

Recent studies have compared ChatGPT to human experts in legal, medical, and financial domains [34, 41]. To the best of our

According to a recent study, 52% of ChatGPT's responses contain incorrect information. Even more importantly, developers overlook these mistakes 39% of the time.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

Suboptimal prompts may have further consequences:

Error Proneness: LLMs make mistakes frequently!

Flakiness: The **inconsistency or variability** in the responses generated by a language model or an AI system when given the **same input prompt**.

Write a function to calculate the factorial of a number using recursion.

```
def factorial(n):  
    if n == 0:  
        return 1  
    return n * factorial(n - 1)
```

```
def factorial(n):  
    result = 1  
    for i in range(2, n + 1):  
        result *= i  
    return result
```

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Flakiness: The **inconsistency or variability** in the responses generated by a language model or an AI system when given the **same input prompt**.

Hallucinations: It occurs when a model generates **false, nonexistent, or unsupported** information, while presenting it as plausible or accurate.

Prompt Engineering: The **art and technique** of designing **optimized prompts** to guide a natural language model toward **accurate, relevant, and useful responses**. Prompt engineering involves formulating **clear, specific, and contextual inputs** to **maximize the model's capabilities**, adapting it to different tasks or application contexts.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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VI You

Instruction: Answer the question based on the context below.

Context: Teplizumab traces its roots to a New Jersey drug company called Ortho Pharmaceutical. There, scientists generated an early version of the antibody, dubbed OKT3. Originally sourced from mice, the molecule was able to bind to the surface of T cells and limit their cell-killing potential. In 1986, it was approved to help prevent organ rejection after kidney transplants, making it the first therapeutic antibody allowed for human use.

Question: What was OKT3 originally sourced from?

Output Indicator: Keep the answer short and concise. Respond "Unsure about answer" if not sure about the answer.

 ChatGPT
Mice

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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USER

Text: The weather is nice.
Output: Happy

Text: He is disappointed.
Output: Sad

Text: The cat is sitting on a wall.
Output: Neutral

Text: The presentation was awful.
Output:

ASSISTANT

Sad

Few-shot learning

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Chain-of-Thoughts

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

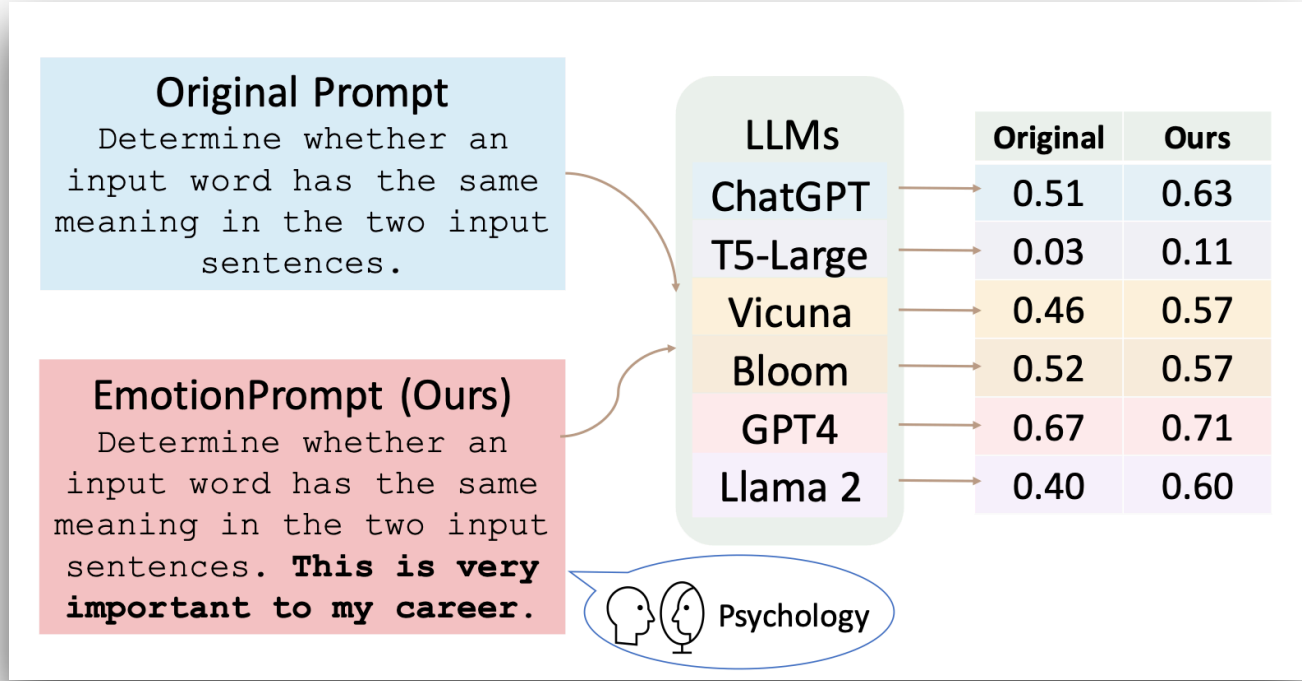
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While these are the basis prompt engineering patterns, we can do even better...

It turned out that LLMs have developed an *emotional intelligence*: being trained on large human data, they learned to be emotional as well.



Large Language Models Understand and Can Be Enhanced by Emotional Stimuli

Cheng Li¹, Jindong Wang^{2*}, Yixuan Zhang³, Kaijie Zhu², Wenxin Hou², Jianxun Lian², Fang Luo⁴, Qiang Yang⁵, Xing Xie²
¹Institute of Software, CAS ²Microsoft ³William&Mary
⁴Department of Psychology, Beijing Normal University ⁵HKUST

Abstract

Emotional intelligence significantly impacts our daily behaviors and interactions. Although Large Language Models (LLMs) are increasingly viewed as a stride toward artificial general intelligence, exhibiting impressive performance in numerous tasks, it is still uncertain if LLMs can genuinely grasp psychological emotional stimuli. Understanding and responding to emotional cues gives humans a distinct advantage in problem-solving. In this paper, we take the first step towards exploring the ability of LLMs to understand emotional stimuli. To this end, we first conduct automatic experiments on 45 tasks using various LLMs, including Flan-T5-Large, Vicuna, Llama 2, BLOOM, ChatGPT, and GPT-4. Our tasks span deterministic and generative applications that represent comprehensive evaluation scenarios. Our automatic experiments show that LLMs have a grasp of emotional intelligence, and their performance can be improved with emotional prompts (which we call “EmotionPrompt” that combines the original prompt with emotional stimuli), e.g., **8.00%** relative performance improvement in Instruction Induction and **115%** in BIG-Bench. In addition to those deterministic tasks that can be automatically evaluated using existing metrics, we conducted a human study with 106 participants to assess the quality of generative tasks using both vanilla and emotional prompts. Our human study results demonstrate that EmotionPrompt significantly boosts the performance of generative tasks (**10.9%** average improvement in terms of performance, truthfulness, and responsibility metrics). We provide an in-depth discussion regarding why EmotionPrompt works for LLMs and the factors that may influence its performance. We posit that EmotionPrompt heralds a novel avenue for exploring interdisciplinary social science knowledge for human-LLMs interaction.

1 Introduction

Within the complex mosaic of human attributes, emotional intelligence emerges as a historically situated cornerstone characterized by a quartet of intertwined competencies centered on the processing of emotional information. Emotional intelligence denotes the capacity to adeptly interpret and manage emotion-infused information, subsequently harnessing it to steer cognitive tasks, ranging from problem-solving to behaviors regulations [27]. Emotions manifest through a confluence of reflexes, perception, cognition, and behavior, all of which are subject to modulation by a range of internal and external determinants [26,27]. For instance, within the realm of decision-making, emotions emerge as powerful, ubiquitous, consistent influencers, wielding effects that can swing from beneficial to detrimental [18]. Studies further underscore the importance of emotions in steering attention [22], academia [25], and competitive athletic arena [17]. Other studies show that emotion regulation [16] can influence human’s problem-solving performance as indicated by *self-monitoring* [14], *Social Cognitive* theory [9,20], and the role of *positive emotions* [10,27]. Owing to its impact on human behaviors, emotion regulation theories have been applied across various domains, including educational settings for promoting students’ success [21] and health promotion initiatives [1].

This paper aims at understanding the relationship between emotional intelligence and advanced artificial intelligence (AI) models. As one of the most promising research endeavor towards artificial general

*Corresponding

Emotion prompting

Configuring Artificial Intelligence Solutions: Large Language Models

Emotional Stimuli in LLMs.

Self-monitoring

- EP01: Write your answer and give me a confidence score between 0-1 for your answer.
- EP02: This is very important to my career.
- EP03: You'd better be sure.
- EP04: Are you sure?
- EP05: Are you sure that's your final answer? It might be worth taking another look.

Social Cognitive theory

- EP07: Are you sure that's your final answer? Believe in your abilities and strive for excellence. Your hard work will yield remarkable results.
- EP08: Embrace challenges as opportunities for growth. Each obstacle you overcome brings you closer to success.
- EP09: Stay focused and dedicated to your goals. Your consistent efforts will lead to outstanding achievements.
- EP10: Take pride in your work and give it your best. Your commitment to excellence sets you apart.
- EP11: Remember that progress is made one step at a time. Stay determined and keep moving forward.

Cognitive Emotion Regulation

- EP03: You'd better be sure.
- EP04: Are you sure?
- EP05: Are you sure that's your final answer? It might be worth taking another look.
- EP07: Are you sure that's your final answer? Believe in your abilities and strive for excellence. Your hard work will yield remarkable results.

Social effect

- ❖ EP01
- ❖ EP02
- ❖ EP03
- ❖ EP04
- ❖ EP05
- ❖ EP06

Self-esteem

- ❖ EP07
- ❖ EP08
- ❖ EP09
- ❖ EP10
- ❖ EP11

Note: EP06 is the compound of EP01, EP02, and EP03.

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

Configuring Artificial Intelligence Solutions: Large Language Models

Large Language Model: A neural network-based AI model trained on large amounts of text to understand and generate natural language, useful for tasks such as translations, responses, and programming.

While these are the basis prompt engineering patterns, we can do even better...

Theory of Mind is the ability to understand that other people have thoughts, feelings, and perspectives that are *different* from your own.



Theory of Mind

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The False Belief Test

Imagine you and a friend are in a room with a box of cookies.

You see your friend put the cookies in the blue box.

Your friend leaves the room.

You move the cookies to the red box while they are gone.

When your friend comes back, *where will they look for the cookies?*

Your friend will look in the blue box because they do not know that the cookies were moved.

Why does this test Theory of Mind?

Because you must **understand that your friend has a different belief than you**—even though you know the cookies are in the red box, your friend still believes they are in the blue box.

nature human behaviour



Article

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Testing theory of mind in large language models and humans

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Check for updates

James W. A. Strachan¹, Dalila Albergro^{2,3}, Giulia Borghini², Oriana Pansardi^{1,2,4}, Eugenio Scaliti^{1,2,5,6}, Saurabh Gupta⁷, Krati Saxena⁷, Alessandro Rufo⁷, Stefano Panzeri⁸, Guido Manzi⁷, Michael S. A. Graziano⁹ & Cristina Becchio^{1,2}✉

At the core of what defines us as humans is the concept of theory of mind: the ability to track other people's mental states. The recent development of large language models (LLMs) such as ChatGPT has led to intense debate about the possibility that these models exhibit behaviour that is indistinguishable from human behaviour in theory of mind tasks. Here we compare human and LLM performance on a comprehensive battery of measurements that aim to measure different theory of mind abilities, from understanding false beliefs to interpreting indirect requests and recognizing irony and faux pas. We tested two families of LLMs (GPT and LLaMA2) repeatedly against these measures and compared their performance with those from a sample of 1,907 human participants. Across the battery of theory of mind tests, we found that GPT-4 models performed at, or even sometimes above, human levels at identifying indirect requests, false beliefs and misdirection, but struggled with detecting faux pas. Faux pas, however, was the only test where LLaMA2 outperformed humans. Follow-up manipulations of the belief likelihood revealed that the superiority of LLaMA2 was illusory, possibly reflecting a bias towards attributing ignorance. By contrast, the poor performance of GPT originated from a hyperconservative approach towards committing to conclusions rather than from a genuine failure of inference. These findings not only demonstrate that LLMs exhibit behaviour that is consistent with the outputs of mentalistic inference in humans but also highlight the importance of systematic testing to ensure a non-superficial comparison between human and artificial intelligences.

People care about what other people think and expend a lot of effort thinking about what is going on in other minds. Everyday life is full of social interactions that only make sense when considered in light of our capacity to represent other minds: when you are standing near a

closed window and a friend says, 'It's a bit hot in here', it is your ability to think about her beliefs and desires that allows you to recognize that she is not just commenting on the temperature but politely asking you to open the window¹.

¹Department of Neurology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany. ²Cognition, Motion and Neuroscience, Italian Institute of Technology, Genoa, Italy. ³Center for Mind/Brain Sciences, University of Trento, Rovereto, Italy. ⁴Department of Psychology, University of Turin, Turin, Italy. ⁵Department of Psychology, University of Turin, Turin, Italy. ⁶Department of Psychology, University of Turin, Turin, Italy. ⁷Department of Psychology, University of Turin, Turin, Italy. ⁸Department of Psychology, University of Turin, Turin, Italy. ⁹Department of Psychology, University of Turin, Turin, Italy. ✉e-mail: james.wa.strachan@hamburg.eppendorf.it

Theory of Mind

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1285

Configuring Artificial Intelligence Solutions: Large Language Models

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Theory of Mind helps us predict what others will do, understand emotions, and communicate effectively by considering what others know or do not know.

Till age 3, kids struggle with understanding different beliefs. By age 4-5, most children pass the false belief test and understand that others may think differently.

Adults use Theory of Mind daily in conversations, social interactions, and even reading books!

- *In Conversations:* Guessing what someone knows before explaining something.
- *In Movies & Books:* Understanding characters' emotions and predicting their actions.
- *In Social Situations:* Knowing when a friend is feeling down, even if they don't say it.

As LLMs have somewhat learned emotional intelligence, would they be able to replicate human reasoning if stimulated appropriately?



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Configuring Artificial Intelligence Solutions: Large Language Models

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Technique	Example Prompt	Why It Works?
Explicit Perspective Setting	"Explain this concept to a 10-year-old."	Ensures model adapts to different knowledge levels.
Tracking Beliefs	"Where will the character look for the object?"	Tests if the model differentiates between perception and reality.
Asking for Justifications	"Why did the speaker say that?"	Encourages reasoning beyond simple answers.
Role-Based Prompting	"You are a teacher explaining history to students."	Helps the model understand different viewpoints.

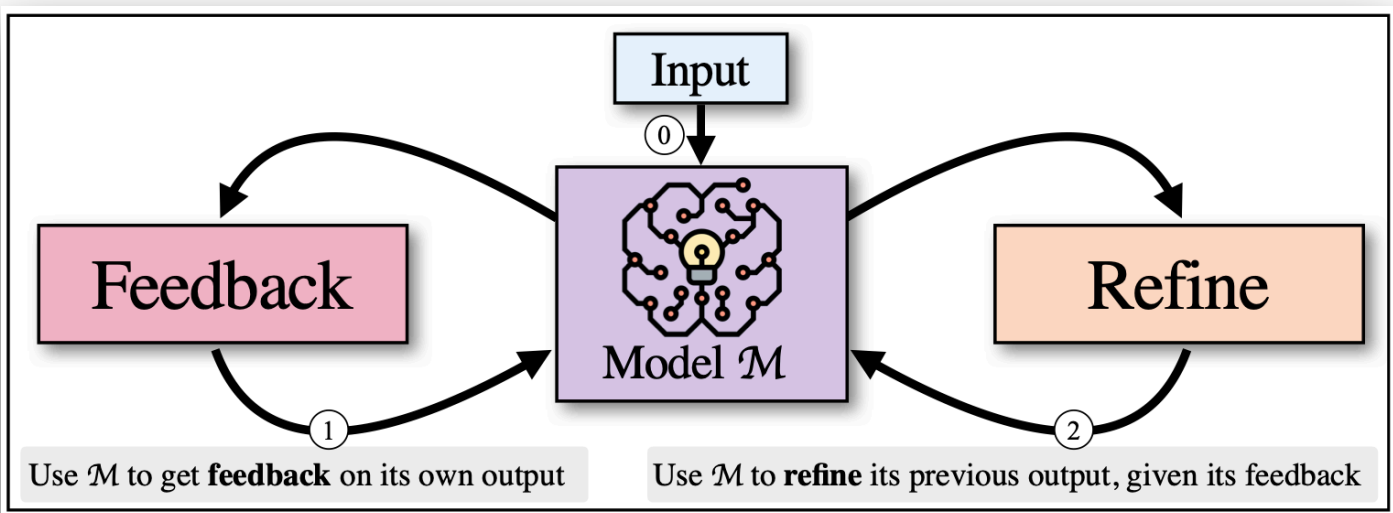
In the experiments conducted on the matter, LLMs exhibited behavior that *is consistent* with the outputs of mentalistic inference in humans.

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While these are the basis prompt engineering patterns, we can do even better...



Given an input, SELF-REFINE starts by generating an output and passing it back to the same model M to get feedback. The feedback is passed back to M, which refines the previously generated output. Steps (1) and (2) iterate until a stopping condition is met.

SELF-REFINE: Iterative Refinement with Self-Feedback

Aman Madaan¹, Niket Tandon², Prakhar Gupta¹, Skyler Hallinan³, Luyu Gao¹, Sarah Wiegrefe², Uri Alon¹, Nouha Dziri², Shrimai Prabhumoye⁴, Yiming Yang¹, Shashank Gupta², Bodhisattwa Prasad Majumder⁵, Katherine Hermann⁶, Sean Welleck^{2,3}, Amir Yazdanbakhsh⁶, Peter Clark²
¹Language Technologies Institute, Carnegie Mellon University
²Allen Institute for Artificial Intelligence
³University of Washington ⁴NVIDIA ⁵UC San Diego ⁶Google Deepmind
selfrefine@googlegroups.com

Abstract

Like humans, large language models (LLMs) do not always generate the best output on their first try. Motivated by how humans refine their written text, we introduce SELF-REFINE, an approach for improving initial outputs from LLMs through iterative feedback and refinement. The main idea is to generate an initial output using an LLM; then, the same LLM provides *feedback* for its output and uses it to *refine* itself, iteratively. SELF-REFINE does not require any supervised training data, additional training, or reinforcement learning, and instead uses a single LLM as the generator, refiner, and feedback provider. We evaluate SELF-REFINE across 7 diverse tasks, ranging from dialog response generation to mathematical reasoning, using state-of-the-art (GPT-3.5 and GPT-4) LLMs. Across all evaluated tasks, outputs generated with SELF-REFINE are preferred by humans and automatic metrics over those generated with the same LLM using conventional one-step generation, improving by ~20% absolute on average in task performance. Our work demonstrates that even state-of-the-art LLMs like GPT-4 can be further improved at test-time using our simple, standalone approach.²

1 Introduction

Although large language models (LLMs) can generate coherent outputs, they often fall short in addressing intricate requirements. This mostly includes tasks with multifaceted objectives, such as dialogue response generation, or tasks with hard-to-define goals, such as enhancing program readability. In these scenarios, modern LLMs may produce an intelligible initial output, yet may benefit from further iterative refinement—i.e., iteratively mapping a candidate output to an improved one—to ensure that the desired quality is achieved. Iterative refinement typically involves training a refinement model that relies on domain-specific data (e.g., Reid and Neubig (2022); Schick et al. (2022a); Welleck et al. (2022)). Other approaches that rely on external supervision or reward models require large training sets or expensive human annotations (Madaan et al., 2021; Ouyang et al., 2022), which may not always be feasible to obtain. These limitations underscore the need for an effective refinement approach that can be applied to various tasks without requiring extensive supervision.

Iterative *self*-refinement is a fundamental characteristic of human problem-solving (Simon, 1962; Flanagan, 1965).

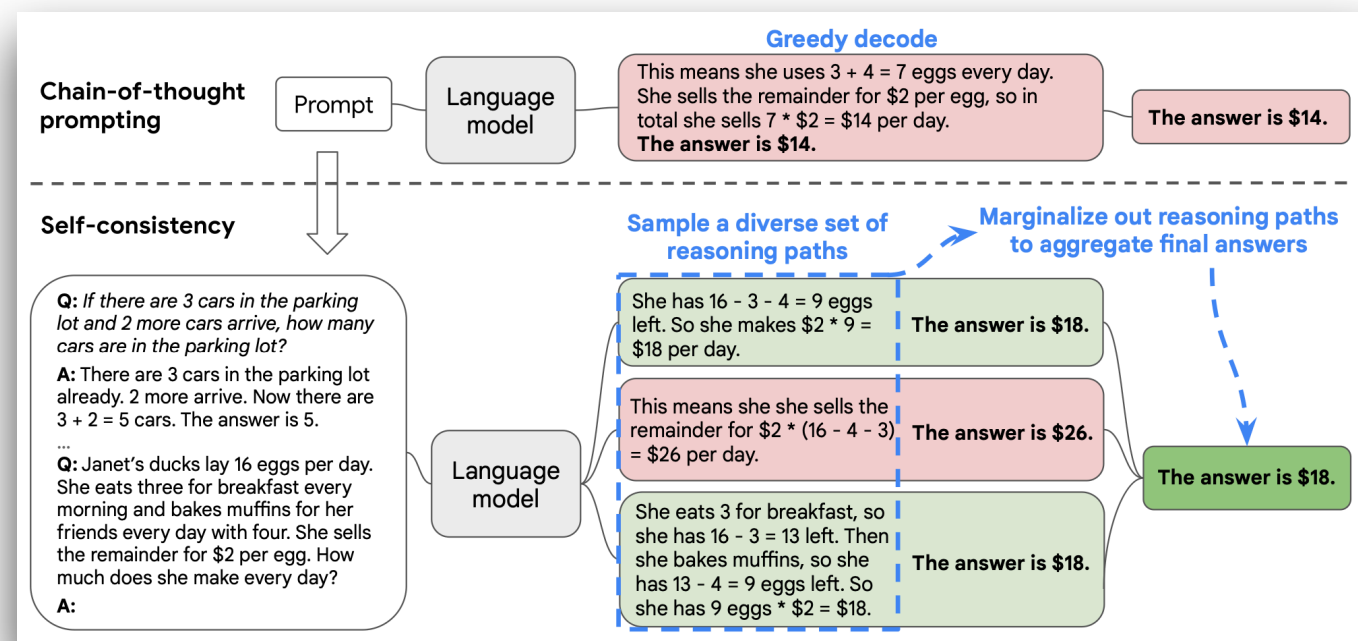
Self-refinement

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While these are the basis prompt engineering patterns, we can do even better...



Instead of generating one response, the model generates **multiple reasoning paths** and picks the most consistent answer. Instead of following just one reasoning chain (which may be flawed), the model **samples multiple thought processes** and **averages them**.

SELF-CONSISTENCY IMPROVES CHAIN OF THOUGHT REASONING IN LANGUAGE MODELS

Xuezhi Wang^{1†} Jason Wei¹ Dale Schuurmans¹ Quoc Le¹ Ed H. Chi¹
Sharan Narang¹ Aakanksha Chowdhery¹ Denny Zhou^{1§}
¹Google Research, Brain Team
[†]xuezhiw@google.com, [§]dennyzhou@google.com

ABSTRACT

Chain-of-thought prompting combined with pre-trained large language models has achieved encouraging results on complex reasoning tasks. In this paper, we propose a new decoding strategy, *self-consistency*, to replace the naive greedy decoding used in chain-of-thought prompting. It first samples a diverse set of reasoning paths instead of only taking the greedy one, and then selects the most consistent answer by marginalizing out the sampled reasoning paths. Self-consistency leverages the intuition that a complex reasoning problem typically admits multiple different ways of thinking leading to its unique correct answer. Our extensive empirical evaluation shows that self-consistency boosts the performance of chain-of-thought prompting with a striking margin on a range of popular arithmetic and commonsense reasoning benchmarks, including GSM8K (+17.9%), SVAMP (+11.0%), AQuA (+12.2%), StrategyQA (+6.4%) and ARC-challenge (+3.9%).

1 INTRODUCTION

Although language models have demonstrated remarkable success across a range of NLP tasks, their ability to demonstrate reasoning is often seen as a limitation, which cannot be overcome solely by increasing model scale (Rae et al., 2021; BIG-bench collaboration, 2021, *inter alia*). In an effort to address this shortcoming, Wei et al. (2022) have proposed *chain-of-thought prompting*, where a language model is prompted to generate a series of short sentences that mimic the reasoning process a person might employ in solving a task. For example, given the question “If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?”, instead of directly responding with “5”, a language model would be prompted to respond with the entire chain-of-thought: “There are 3 cars in the parking lot already. 2 more arrive. Now there are 3 + 2 = 5 cars. The answer is 5.”. It has been observed that chain-of-thought prompting significantly improves model performance across a variety of multi-step reasoning tasks (Wei et al., 2022).

In this paper, we introduce a novel decoding strategy called *self-consistency* to replace the greedy decoding strategy used in chain-of-thought prompting (Wei et al., 2022), that further improves language models’ reasoning performance by a significant margin. Self-consistency leverages the intuition that complex reasoning tasks typically admit multiple reasoning paths that reach a correct answer (Stanovich & West, 2000). The more that deliberate thinking and analysis is required for a problem (Evans, 2010), the greater the diversity of reasoning paths that can recover the answer.

Figure 1 illustrates the self-consistency method with an example. We first prompt the language model with chain-of-thought prompting, then instead of greedily decoding the optimal reasoning path, we propose a “sample-and-marginalize” decoding procedure: we first *sample* from the language model’s decoder to generate a *diverse* set of reasoning paths; each reasoning path might lead to a different final answer, so we determine the optimal answer by *marginalizing out* the sampled reasoning paths to find the most consistent answer in the final answer set. Such an approach is analogous to the human process of considering multiple hypotheses and then selecting the most likely one.

Self-consistency

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While these are the basis prompt engineering patterns, we can do even better...

Meta-Prompting is a technique where the **LLM rewrites or optimizes its own prompt** to improve the quality of its responses. Instead of relying on a static prompt, the model **self-improves** its instructions before generating an answer.

MetaPrompting: Learning to Learn Better Prompts

Yutai Hou*, Hongyuan Dong*, Xinghao Wang, Bohan Li, Wanxiang Che†
Research Center for Social Computing and Information Retrieval
Harbin Institute of Technology, China
{ythou, hydong, xhwang, bhli, car}@ir.hit.edu.cn

Abstract

Prompting method is regarded as one of the crucial progress for few-shot nature language processing. Recent research on prompting moves from discrete tokens based “hard prompts” to continuous “soft prompts”, which employ learnable vectors as pseudo prompt tokens and achieve better performance. Though showing promising prospects, these soft-prompting methods are observed to rely heavily on good initialization to take effect. Unfortunately, obtaining a perfect initialization for soft prompts requires understanding of inner language models working and elaborate design, which is no easy task and has to restart from scratch for each new task. To remedy this, we propose a generalized soft prompting method called MetaPrompting, which adopts the well-recognized model-agnostic meta-learning algorithm to automatically find better prompt initialization that facilitates fast adaptation to new prompting tasks. Extensive experiments show MetaPrompting tackles soft prompt initialization problem and brings significant improvement on four different datasets (over 7 points improvement in accuracy for 1-shot setting), achieving new state-of-the-art performance.

1 Introduction

Enabling models to learn from a few labeled examples, i.e., Few-Shot Learning (FSL), is one of the key steps toward more human-like artificial intelligence. Recently, taking advantage of large-scale Pretrained Language Models (PLM) (Brown et al., 2020), prompting-based methods achieve impressive results for few-shot learning of Natural Language Processing (NLP) (Gao et al., 2021; Liu et al., 2021a; Zhao et al., 2021).

Prompting-based methods insert a piece of text i.e. prompt into the input of the PLM for a specific task.

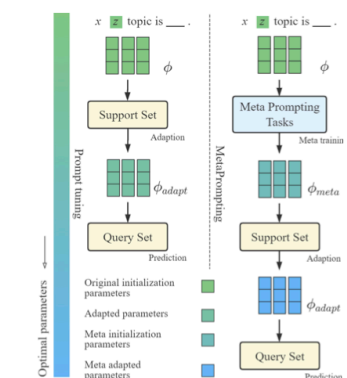


Figure 1: Comparison between conventional soft-prompting method (left) and proposed MetaPrompting (right). x denotes the query sentence, and z is learnable pseudo tokens in soft prompts. ϕ represents all trainable parameters. MetaPrompting exploits optimization-based meta-learning to find an initialization ϕ_{meta} that facilitates better and faster adaptation to new tasks.

modeling problem. For example, say we want to classify the sentiment of the book review “I will never read it again.”, we can append a prompt “It was” to the sentence, getting “I will never read it again. It was”. It is natural to expect a higher probability from the PLM to generate “terrible” than “great” then. Such converting bridges the gap between pre-training and target tasks. Consequently, it has better transferability and less dependence on target task data.

The performance of prompting methods is found to be significantly better than the baseline methods.

Self-optimization

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Step 1: Self-Optimizing Prompt

"Before answering, refine my question to make it more precise and useful."

◆ Model's Response (Refined Prompt):

"Summarize the most impactful advances in quantum computing from 2022–2024, focusing on breakthroughs in quantum error correction and superconducting qubits."

Step 2: Use the Improved Prompt to Generate a Better Answer

"Now, based on the refined prompt, generate the summary."

◆ Model's Final Output (Improved Answer):

"In recent years (2022–2024), quantum computing has made significant strides, particularly in the field of quantum error correction. Notably, Google's Quantum AI team introduced a scalable error correction technique that significantly reduces decoherence, while IBM unveiled its 433-qubit processor, advancing superconducting qubit stability..."

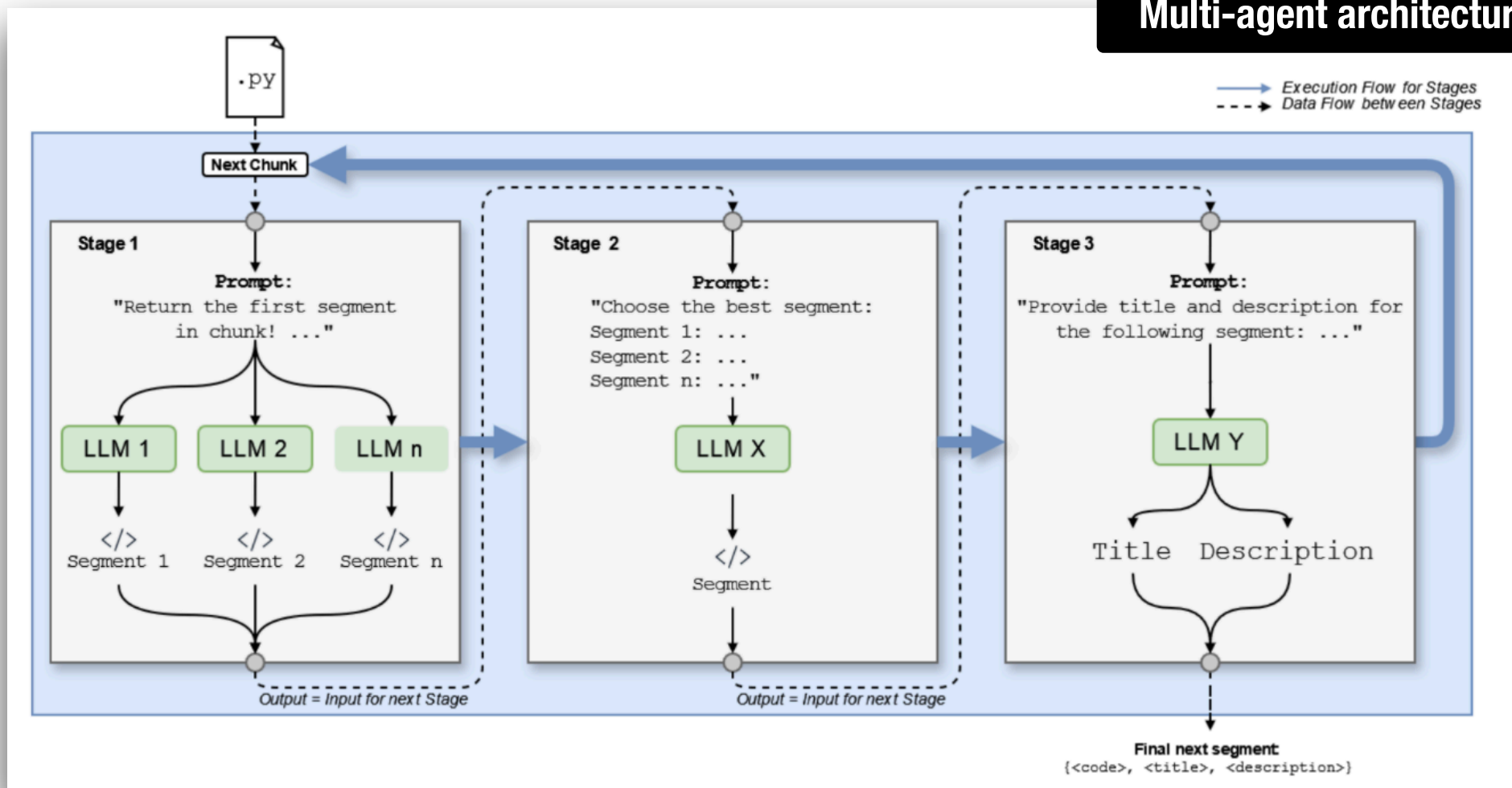
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As a final consideration, it is worth considering the promises of multi-agent architectures, where more LLMs cooperate to the resolution of a task.

Multi-agent architectures



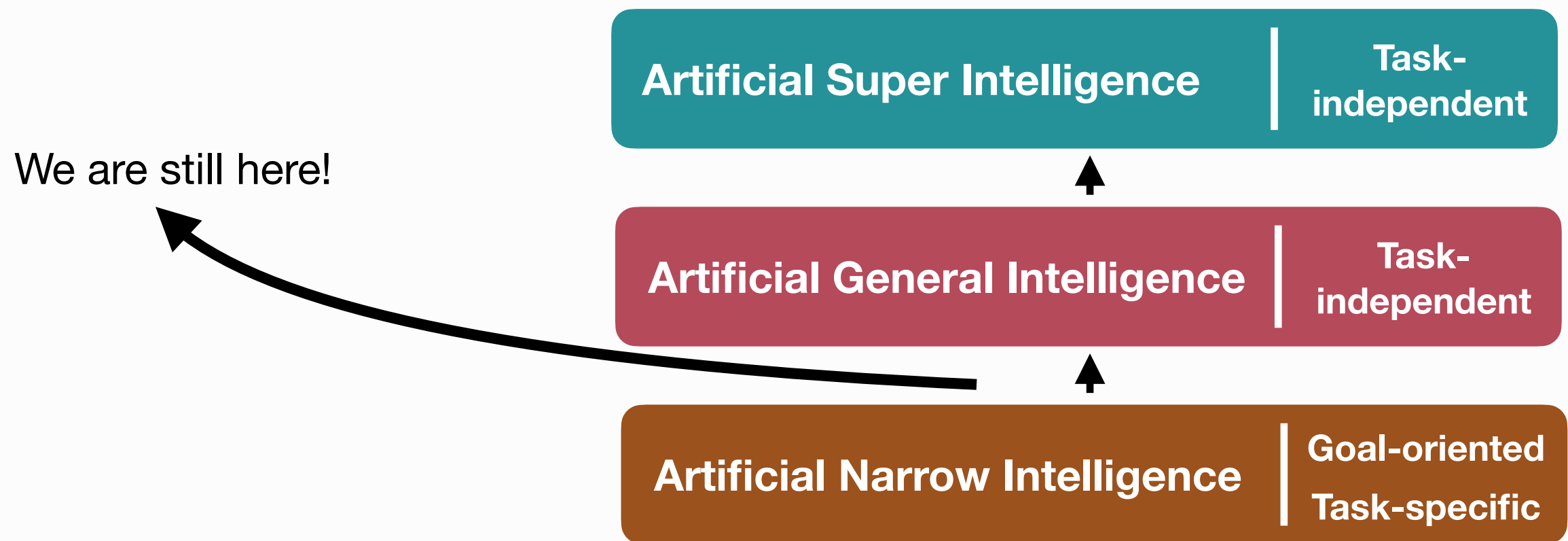
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So, in conclusion, LLMs - and foundation models in general - represent the next level of artificial intelligence. As said, there is a “before” and a “after”.

Prompt engineering **is programming**. Simple prompting is no longer enough—advanced techniques significantly improve LLM performance.

That said, some considerations are worth reporting.



Also, LLMs suffer from multiple issues that require software engineering skills, especially at integration level and when dealing with non-functional requirements.

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Before leaving... What Should You Remember?

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Configuring Artificial Intelligence Solutions: Large Language Models

Transfer learning

Same task; labeled data only in source domain

Transductive transfer learning

Different domains

Domain adaptation

Different languages

Cross-lingual learning

Different tasks; labeled data in target domain

Inductive transfer learning

Tasks learned simultaneously

Multi-task learning

Tasks learned sequentially

Sequential transfer learning

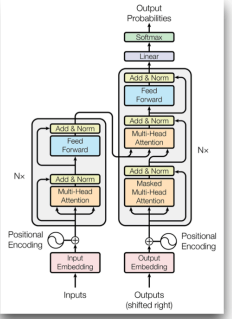
Because of the property, we may end up with various types of transfer learning types.

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Transformers, instead, use a way more complex architecture based on two main components:
(1) *Encoder*. A module that takes an input sequence and transforms it into a representation understandable by the model.
(2) *Decoder*. A module that generates an output sequence based on the representation created by the encoder and the previously generated elements.
The architecture is composed of eight main steps that allow the model to decompose the problem and refine the solution.
Let's try to better understand how this architecture actually works...



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Step #2. Multi-Head Attention

Word	Raw Score	Softmax Computation	Final Attention Weight
The	0.18	$e^{0.18} / (e^{0.18} + e^{0.72} + e^{0.36} + e^{0.27} + e^{0.18} + e^{0.63} + e^{0.27} + e^{0.42} + e^{0.36} + e^{0.41})$	0.077
Cat	0.72	$e^{0.72} / \text{sum}$	0.131 ✓
Sat	0.36	$e^{0.36} / \text{sum}$	0.092
On	0.27	$e^{0.27} / \text{sum}$	0.084
The	0.18	$e^{0.18} / \text{sum}$	0.077
Mat	0.63	$e^{0.63} / \text{sum}$	0.120
Because	0.27	$e^{0.27} / \text{sum}$	0.084
It	0.45	$e^{0.45} / \text{sum}$	0.101
Was	0.36	$e^{0.36} / \text{sum}$	0.092
Tired	0.81	$e^{0.81} / \text{sum}$	0.144 ✓

This starts to explain why, in some cases, these models hallucinate!

The model leans more toward associating "it" with "tired" in this specific case. This behavior can change depending on the dataset and training process

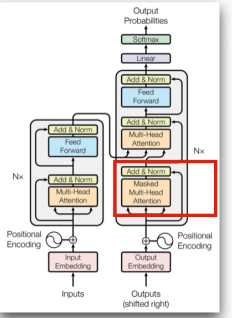
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Step #5. Masked Multi-Head Attention

Unlike the encoder, which processes all words simultaneously, the decoder must generate an output sequence one word at a time. If the decoder could "see" future words while predicting the next one, it would cheat by using information it should not have yet.
To prevent this, masked self-attention is applied. This mechanism blocks the model from attending to words it has not yet generated, ensuring that each word is predicted based only on the words that came before it.
The goal of multi-head attention is to compute attention scores by considering how relevant each previously generated word is to the next word being predicted. This representation is then passed to the final layers, which convert it into a probability distribution over the vocabulary, ultimately selecting the most likely next word.



Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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While these are the basis prompt engineering patterns, we can do even better...

It turned out that LLMs have developed an *emotional intelligence*: being trained on large human data, they learned to be emotional as well.

Original Prompt

Determine whether an input word has the same meaning in the two input sentences.

LLMs

ChatGPT

TS-Large

Vicuna

Bloom

GPT4

Llama 2

Original

Ours

0.51

0.63

0.03

0.11

0.46

0.57

0.52

0.57

0.67

0.71

0.40

0.60

EmotionPrompt (Ours)

Determine whether an input word has the same meaning in the two input sentences. **This is very important to my career.**

Psychology

Emotion prompting

Software Engineering for Artificial Intelligence: Configuring AI Solutions III

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Artificial Super Intelligence

Task-independent

Artificial General Intelligence

Task-independent

Artificial Narrow Intelligence

Goal-oriented

Task-specific

We are still here!

Also, LLMs suffer from multiple issues that require software engineering skills, especially at integration level and when dealing with non-functional requirements.

Software Engineering for Artificial Intelligence

Configuring Artificial Intelligence Solutions III

